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# Mouse Locus Coeruleus Noradrenergic Neurons: Structure, Function, Location, Markers and Transcriptomics

## Table of Contents


Introduction

Anatomical location and gross organization

Cellular structure and morphological/electrophysiological features

Molecular markers and neurochemical identity

Functions and projection-defined heterogeneity

Developmental origin and subpopulation organization

Transcriptomics datasets and molecular profiling


## Introduction

*The mouse locus coeruleus (LC) is a small brainstem nucleus that contains the main group of noradrenergic neurons in the CNS. Even though it is compact, it sends broad signals across the brain and helps control attention, learning, stress, pain, memory, and sleep. (2 sources)*

In the mouse, the locus coeruleus is a relatively small nucleus in the brainstem, but it is the primary location of noradrenergic neurons and the main source of norepinephrine for the central nervous system. (Tanguay et al., 2023) (Yaman et al., 2024) This compact nucleus has outsized influence because LC neurons project widely and can shape activity across many target regions, including cortex, hippocampus, and thalamus. (Yaman et al., 2024) As a result, LC noradrenergic neurons are central to broad brain-state control and are linked to attention, learning, stress responses, pain modulation, memory, and sleep. (Yaman et al., 2024) Current work also stresses that the LC is not a uniform cell group: its neurons show anatomical and functional heterogeneity, which is important for understanding how different LC subpopulations support distinct forms of neuromodulation. (Yaman et al., 2024)

### Evidence

**(Tanguay et al., 2023)**

Shining light on the noradrenergic system

> *"the relatively small size and the position in the brain of the locus coeruleus (LC), the primary location for noradrenergic neurons."*

**(Yaman et al., 2024)**

The locus coeruleus in aging and neurodegenerative diseases

> *"The locus coeruleus (LC), a prominent neuromelanin-containing nucleus, plays a critical role in the central nervous system by serving as the main source of norepinephrine"*
>
> *".the LC influences various brain functions, including attention, learning, stress responses, pain modulation, memory, and sleep"*
>
> *".This review explores the anatomy, morphology, and neurochemistry of LC neurons, emphasizing their projections and interactions with multiple brain regions such as the cortex, hippocampus, and thalamus"*
>
> *".Understanding the anatomical and functional heterogeneity of LC neurons provides insights into their crucial role in*

*neuromodulation"*

# Anatomical location and gross organization

*In the mouse, the LC is a small paired nucleus in the dorsal/rostral pons, right next to the fourth ventricle. It is very compact and packed with mostly noradrenergic neurons, but these neurons send axons widely across the brain and down to the spinal cord. (17 sources)*

Anatomically, the mouse locus coeruleus is a **bilateral brainstem nucleus in the dorsal pontine tegmentum**, classically described along the **lateral floor or margin of the fourth ventricle** in the **rostral/posterior pons**. It is often identified in coronal sections near **bregma ~−5.4 mm**, where it lies close to structures such as the medial parabrachial nucleus, Barrington's nucleus, and laterodorsal tegmental nucleus, with the fourth ventricle as a key landmark. (Carvalho et al., 2014) (Liu et al., 2013) (Drenan et al., 2008) (Lutsenko et al., 2019) (Kelly et al., 2023) (Lee et al., 2024)

Grossly, the LC is **small but dense**. Several mouse-focused sources describe it as containing about **~1500 neurons per side** (roughly ~3000 total), although some papers report ~1500 total or use rodent-wide estimates near 300–500 µm across** in common section planes, which helps explain why it can be hard to localize during dissection or slice work. (Carvalho et al., 2014) (Lutsenko et al., 2019) (Gargano et al., 2020) (Cao et al., 2020) (Schmidt et al., 2019) (Gonye et al., 2023) (Zhou et al., 2024)

In section view, the LC has a recognizable **paired "horn-like" appearance** at the edges of the ventricle, with a **densely packed nuclear core** that contains the highest density of noradrenergic neurons. Recent work also emphasizes that this compact core sits within a **more heterogeneous peri-LC surround**, where non-noradrenergic cells are intermingled. (Lutsenko et al., 2019) (Zouridis et al., 2024)

Although best known as the **A6 noradrenergic group**, the LC is often considered part of a broader **locus coeruleus complex** in the rostral hindbrain that includes nearby noradrenergic populations such as **A4, A5, and A7/subcoeruleus-related groups**. In mammals, these neurons occupy the **periventricular gray matter and adjacent parvicellular reticular regions**, and in murid rodents the A6 group is especially dense. (Manger et al., 2021) (Robertson et al., 2013)

A notable organizational feature is that the LC is **compact locally but broad globally**: the somata are clustered tightly together, with many neighboring cells separated by only tens of micrometers, yet their axons project across almost the entire neuraxis. About a quarter of LC noradrenergic neurons have another LC neuron within **<25 µm**, consistent with the idea of a tightly packed local ensemble. At the systems level, LC neurons innervate **forebrain, thalamus, hypothalamus, amygdala, hippocampus, cerebellum, brainstem, and spinal cord**, making the LC the major source of central norepinephrine despite its small size. (Baral et al., 2022) (Carvalho et al., 2014) (Liu et al., 2013) (Gargano et al., 2020) (Alam et al., 2023) (Lee et al., 2024)

For practical mouse anatomy, investigators often use **appearance of the fourth ventricle**, disappearance of the cerebral aqueduct, and **TH or DBH staining** to confirm LC position. These markers reliably outline the bilateral noradrenergic cell columns and are especially useful because the mouse LC lacks strong gross landmarks and is so small. (Cao et al., 2020) (Schmidt et al., 2019) (Drenan et al., 2008) (Holland et al., 2021) (Lee et al., 2024)

## Evidence

**(Carvalho et al., 2014)**

Neurochemical and electrical modulation of the locus coeruleus: contribution to CO2drive to breathe

*"The locus coeruleus (LC) is a dorsal pontine region, situated bilaterally on the floor of the fourth ventricle. It is considered to be the major source of noradrenergic innervation in the brain"*

*". The LC region is a densely packed nucleus, comprising about 1500 neurons per side (Aston- (Aston-Jones et al., 1984)"*

*". In addition to NA, the cell bodies also secret several neuropeptides (Sutin et al., 1991), among which are vasopressin, somatostatin, neuropeptide Y, enkephalin, neurotensin, CRH, and galanin (for review, see (Olpe et al., 1991)"*

*". LC NA neurons projects broadly throughout the neuraxis, from spinal cord to neocortex (Swanson and Hartman, 1975;Foote et al., 1983;Berridge and Waterhouse, 2003). In fact, it is estimated that  50%of all the noradrenergic projections in the central nervous system (CNS) originate in the LC which are directed toward the forebrain, cerebellum, brainstem and spinal cord (Aston- Jones et al., 1995;Berridge and Waterhouse, 2003). LC is the primary source of an extensive, yet regionally specialized, noradrenergic innervation of the forebrain (Berridge and Waterhouse, 2003) and it is considered a major wakefulness promoting nucleus with activation of the LC resulting in an increase in EEG signs of alertness (Samuels and Szabadi, 2008)."*

### (Liu et al., 2013)

Degenerative alterations in noradrenergic neurons of the locus coeruleus in Alzheimer's disease

*"The locus coeruleus is located in the posterior area of the rostral pons in the lateral floor of the fourth ventricle. It is the major site to secrete the catecholamine neurotransmitter, noradrenaline, and is therefore also known as a noradrenergic cell group (Bouret et al., 2005) . The locus coeruleus has over 50% of the noradrenergic neurons in the entire central nervous system, and is the principal site for brain synthesis of noradrenaline. The nucleus has widespread projections that innervate the spinal cord, cerebellum, hypothalamus, thalamic relay nuclei, amygdala, basal telencephalon, and cortex"*

*". Noradrenaline can regulate the regional microenvironment, and functions both as a neurotransmitter and as an endogenous anti-inflammatory factor (Feinstein et al., 2002)(Heneka et al., 2006)"*

*". Noradrenergic neurons can be visualized by tyrosine hydroxylase, a marker of catecholaminergic neurons"*

*". Immunohistochemistry for tyrosine hydroxylase (a marker of catecholaminergic neurons in the locus coeruleus) revealed that APP/PS1 mice had 23% fewer cells in the locus coeruleus compared with aged wild-type mice. APP/PS1 mice also had increased numbers of cell bodies of neurons positive for tyrosine hydroxylase, but fewer tyrosine hydroxylase-positive fibers, which were also short, thick and broken. Quantitative analysis using unbiased stereology showed a significant age-related increase in the mean volume of tyrosine droxylase-positive neurons in aged APP/PS1 mice compared with young APP/PS1 mice. Moreover, the mean volume of tyrosine hydroxylase-positive neurons was positively correlated with the total volume of the locus coeruleus"*

*". We studied the morphology of noradrenergic neurons in the locus coeruleus using tyrosine hydroxylase immunohistochemistry as a specific marker of catecholaminergic neurons, and quantitatively analyzed changes in the number of tyrosine hydroxylase-positive neurons and fibers using unbiased stereology."*

### (Drenan et al., 2008)

In Vivo Activation of Midbrain Dopamine Neurons via Sensitized, High-Affinity α6* Nicotinic Acetylcholine Receptors

*"(A) Diagram of coronal sections (bregma À5.4 mm) containing locus coeruleus (Franklin and Paxinos, 2008). (B) Tyrosine hydroxylase (TH) staining of NE neurons in coronal slices. Coronal slices (200 mm; bregma À5.4 mm) from an a6 L9 0 S mouse were fixed and stained for TH as described in Supplemental Experimental Procedures. The boxed area from (A) is shown. Scale bar: 100 mm. MPB, medial parabrachial nucleus; LC, locus coeruleus; Bar, Barrington's nucleus; LDTg, laterodorsal tegmental nucleus; 4V, fouth ventricle. (C) Electrophysiological identification of LC neurons. LC neurons are identified by (Ci) pacemaker firing at 1-2 Hz, (Cii) lack of I h , and (Ciii) lack of membrane potential "sag" for hyperpolarizing current pulses (responses shown for injection of À80, À40, and +20 pA)"*

*".(D) Hypersensitive nicotinic responses in LC neurons. Patch-clamp recordings were taken from LC neurons in coronal slices from WT, a4 L9 0 A , and a6 L9 0 S mice. Neurons were voltage clamped at À60 mV, and nicotine at the indicated concentration was locally applied (arrow) to the cell body (250 ms)."*

### (Lutsenko et al., 2019)

COPPER AND THE BRAIN NORADRENERGIC SYSTEM

*"Locus coeruleus (LC) represents a cluster of approximately 1500 neurons located in the pons (pontine tegmentum) region near the 4th ventricle (Fig. 4, upper panel) (Grimm et al., 2004)"*

*".The noradrenergic neurons are concentrated in the LC. Although LC is small, it sends projections throughout the brain and represents the major source of norepinephrine in the CNS. The LC neurons are responsible for modulation of arousal, attention, and vigilance; they accomplish this function through regulation of target regions in a forebrain (Chamberlain et al., 2013)(Sara, 2009) (Szabadi, 2013). In the coronal sections of the brain (Fig. 4, lower panel), LC has a characteristic "horn-like" appearance at the edges of the ventricle and can be easily identified by immunostaining for the Cu-dependent enzyme dopamine-β-hydroxylase, DBH, or tyrosine hydroxylase, which is an established marker of noradrenergic neurons (Fig. 4, lower panel)."*

### (Kelly et al., 2023)

Subunit-specific expression and function of AMPA receptors in the mouse locus coeruleus

*"The locus coeruleus (LC) is a collection of neurons, located bilaterally within the dorsal pons, adjacent to the fourth ventricle (Dahlstroem et al., 1964). LC neurons are distinguished from surrounding structures by their production of the neurotransmitter noradrenaline (NA), which in turn facilitates the brain-wide modulatory role of this nucleus (Berridge & Waterhouse, 2003). By projecting to almost the entire brain, the LC-NA system modulates ongoing brain activity, thereby influencing some of the most salient brain functions, including cognition (Usher et al., 1999), memory (Sara, 2015), arousal (Carter et al., 2010)), mood (McCall et al., 2015) and the central limb of the stress response (Valentino et al., 2008). In addition, the LC also modulates the physiology of various peripheral systems including the control of respiration (Carvalho et al., 2014)(Gargaglioni et al., 2010) and cardiac function (Biancardi et al., 2014)(Patrone et al., 2014)(Yao et al., 2005)"*

*".Central to LC function, in both health and disease, is the spontaneous firing rate (FR) of the principal neurons which determine the pattern and amount of NA release"*

*".In the awake state and in in vitro preparations, LC neurons characteristically fire action potentials spontaneously, with a frequency ranging from ~1 to 5 Hz (Berridge & Waterhouse, 2003)"*

*".GluA1 immunoreactive clusters were associated with puncta immunoreactive for VGLUT2 on somata, and VGLUT1 on distal dendrites. GluA4 was associated with these synaptic markers only in the distal dendrites. No specific signal was detected for the GluA2-3 subunits."*

### (Lee et al., 2024)

Sex Differences in Single Neuron Function and Proteomics Profiles Examined by Patch-Clamp and Mass Spectrometry in the Locus Coeruleus of the Adult Mouse

*"A1 noradrenergic neurons in the medulla, A2 neurons in the nucleus tractus solitarius and A6 neurons in the LC project to many areas in the CNS, including the cortex, hippocampus, thalamus, and spinal cord 18 . The LC, a small nucleus located on either side of the fourth ventricle, contains the largest cluster of noradrenergic neurons in the CNS. LC neurons project to the cortex, hippocampus, amygdala, hypothalamus, cerebellum, and spinal cord 18 and modulate physiological processes, such as arousal, attention, cognition, anxiety, and responses to stress 18-21"*

*"We identified LC noradrenergic neurons in the pons by their location along the lateral margin of the fourth ventricle 28, 29 and by expression of tyrosine hydroxylase (TH) 26, a key enzyme involved in catecholamine synthesis. We also labeled LC neurons with anti-dopamine beta-hydroxylase (DBH) antibodies, an enzyme that converts dopamine to noradrenaline and is a marker of noradrenergic neurons 26, 30."*

### (Gargano et al., 2020)

Lack of Cannabinoid Receptor Type-1 Leads to Enhanced Age-Related Neuronal Loss in the Locus Coeruleus

*"The locus coeruleus (LC), or "blue spot", is a small nucleus located in the pons of the brainstem, lateral to the IVth ventricle. It projects throughout the neuroaxis and represents the major source of noradrenaline (NE) in the central nervous system. Although the LC consists of a relatively low number of noradrenergic neurons (45-50 thousand in a normal healthy young adult human (Sharma et al., 2010) and approximately 1500 in adult mice (Sturrock et al., 1985)), its projecting area is extremely wide; noradrenergic terminals are abundantly present in the cortex, mainly in the somatosensory and motor cortex (Gaspar et al., 1989), hippocampus and amygdala (Gaspar et al., 1989)(Sadikot et al., 1990)(Radley et al., 2008), hypothalamus (Ginsberg et al., 1993) and in the brainstem itself (Westlund et al., 1980). The LC also receives a large number of afferents, mainly from the cortex, amygdala [8] (Pammer et al., 1990) and from the spinal cord (Westlund et al., 1996)"*

*".Electrophysiological studies revealed that LC noradrenergic neurons have a peculiar double firing mode; they are able to switch between tonic or phasic mode, regulating different behavioral states. A tonic firing is related to a "low attention" mode, while the switch to the phasic firing happens in response to a relevant "focus-demanding" stimuli (Aston-Jones et al., 2005). These studies suggested that LC may play a role in modulating numerous cognitive functions, including attention, memory as well as in the regulation of sleep-wake states and stress response (Benarroch, 2017). Under physiological conditions, LC neurons respond to external and internal sensory stimuli and this response influences learning and memory processes (Uematsu et al., 2015)."*

### (Cao et al., 2020)

Fast Localization and Sectioning of Mouse Locus Coeruleus

*"LC is a multifunctional nucleus which is also the source of norepinephrine in the brain"*

*". Locus coeruleus-(LC-) related researches are important parts of neuroscience because it is not only a key hub of noradrenergic neurons, but also the only (or most important) source of norepinephrine in the brain [1]. LC projects into the brain to participate in the regulation of circadian rhythm [2], stress [3], memory and cognition [4], general anesthesia [5], and so on. LC also innervates the spinal cord to regulate physiological and pathological pain [6]"*

*".mouse LC is small (about 500 μm in sagittal plane) [11] and lacks obvious localization landmarks"*

*"the intact brains of adult mice were placed on a horizontal surface and vertically cut along the posterior margin of the bilateral*

*cerebral cortex. In the coronal cutting plane, the aqueduct of midbrain can be seen easily with the naked eyes. After embedding the cerebellum part with optimal cutting temperature (OCT) compound, coronal brain slices were cut from the cutting plane, within 1 mm, the aqueduct of midbrain disappeared and the fourth ventricle appeared, then the brain slices contained LC and were collected. From the first collection, at ~200 μm, the noradrenergic neurons' most enriched brain slices can be collected. The tyrosine hydroxylase immunofluorescence staining confirmed that the localization of LC with this method is accurate and the noradrenergic neuron most abundant slices can be determined with this method."*

**(Schmidt et al., 2019)**

Localization of the Locus Coeruleus in the Mouse Brain

*"The locus coeruleus (LC) is an important region in the brainstem and a major site of norepinephrine (NE) production (Robertson et al., 2013) . The LC sends projections throughout the brain (Kobayashi et al., 1975) to the cortex, the hippocampus and the cerebellum (Olson et al., 1971) and regulates major physiological processes, including circadian rhythm (Costa et al., 2018)(Semba et al., 1984) , attention and memory (Takeuchi et al., 2016) , stress (Korf et al., 1973) , cognitive processes (Sara et al., 1991) , and emotion 9,(File et al., 1979)"*

*"Due to their small size, the mouse LC has an average diameter of ~300 μm, leading to difficulty locating the structure"*

*"Here, we describe a step-by-step immunohistochemistry-based protocol to localize the LC in the mouse brain. Dopamine-β-hydroxylase (DBH), and alternatively, tyrosine hydroxylase (TH), both enzymes highly expressed in the LC, are used as immunohistochemical markers in brain slices."*

**(Gonye et al., 2023)**

Criteria for central respiratory chemoreceptors: experimental evidence supporting current candidate cell groups

*"The locus coeruleus (LC) is a brainstem structure located in the rostral pons, lateral and ventral to the fourth ventricle; it comprises ~3000 noradrenergic neurons in mouse or rat, providing the primary noradrenergic innervation throughout the central nervous system (Loizou, 1969)Schwarz and Luo, 2015;(Liu et al., 2020)(McKinney et al., 2023). Its activity is tightly correlated to arousal levels and stress. Neurons within the LC are electrically coupled via gap junctions, particularly in their dendritic processes, and they exhibit a steady pacemaker-like baseline firing pattern with a pronounced subthreshold oscillation (Ishimatsu et al., 1996)(Oyamada et al., 1999)(Ballantyne et al., 2004)"*

*".There is considerable evidence that LC neurons can influence respiration and display intrinsic CO 2 /H + sensitivity"*

**(Zhou et al., 2024)**

Progressive noradrenergic degeneration and motor cortical dysfunction in Parkinson's disease

*"Locus coeruleus (LC) is a bilateral structure situated in the upper dorsolateral pontine tegmentum. LC neurons synthesize and release norepinephrine (NE, also known as noradrenaline), which is essential for regulating brain function. LC of both hemispheres comprises about 50,000 cells in humans and around 3000 cells in rodents [1](Swanson, 1976)(Baker et al., 2004)(German et al., 1988). The LC-NE neurons project to broad brain regions (e.g., the cerebral cortex, the thalamus, the amygdala, the hippocampus, the cerebellum, and the medulla) and play a pivotal role in regulating normal brain function and behavioral states, including arousal, attention, cognition, and stress responses. Noradrenergic receptors (ARs) can be categorized into 3 subtypes, including α1, α2, and β receptors, which are G-protein coupled receptors (GPCRs). While α1 and β receptors are coupled with G q and G s proteins, respectively, α2 receptors are coupled with G i protein and the subsequent signaling cascades (Perez, 2020). Particularly, α2 receptors are expressed highly at noradrenergic axon terminals, where they can act as autoreceptors to regulate presynaptic NE release. Thus, NE has a higher affinity for α2 receptors than α1 and β receptors."*

**(Zouridis et al., 2024)**

Juxtacellular recordings from identified neurons in the mouse locus coeruleus

*"The locus coeruleus (LC) is the primary source of noradrenergic transmission in the mammalian central nervous system. This small pontine nucleus consists of a densely packed nuclear core—which contains the highest density of noradrenergic neurons—embedded within a heterogeneous surround of non-noradrenergic cells"*

*". We found that noradrenergic neurons (identified by tyrosine hydroxylase, TH, expression; TH-positive) and intermingled putatively non-noradrenergic (TH-negative) cells displayed similar morphologies and responded to foot shock stimuli with excitatory responses; however, on average, TH-positive neurons exhibited more prominent foot shock responses and post-activation firing suppression. The two cell classes also displayed different spontaneous firing rates, spike waveforms and temporal spiking properties."*

**(Manger et al., 2021)**

The Mammalian Locus Coeruleus Complex—Consistencies and Variances in Nuclear Organization

*"These noradrenergic neurons have been shown to be derived from rhombomeres 1-5 in the developing mouse brain (Robertson et*

*al., 2013)"*

*"in mammals, with one known exception (Imam et al., 2018), the noradrenergic neurons of the locus coeruleus complex are found within the periventricular grey matter and the parvicellular reticular nucleus of the rostral hindbrain"*

*"the locus coeruleus complex is comprised of several nuclei"*

*"Descriptions of the nuclear parcellation of the locus coeruleus complex have been provided in approximately 80 mammal species spanning the phylogenetic breadth of this class. Within the mammalian rostral hindbrain, noradrenergic neurons (revealed with tyrosine hydroxylase and dopamine-ß-hydroxylase immunohistochemistry) have been observed within the periventricular grey matter (A4 and A6 nuclei) and parvicellular reticular nucleus (A5 and A7 nuclei)"*

*"In most mammals, the A6 is comprised of a moderate density of neurons, but in Murid rodents, primates, and megachiropteran bats, the A6 exhibits a very high density of neurons."*

### (Robertson et al., 2013)

Developmental origins of central norepinephrine neuron diversity

*"Central norepinephrine-producing neurons comprise a diverse population of cells differing in anatomical location, connectivity, function and response to disease and environmental insult. The mechanisms that generate this diversity are unknown. Here we elucidate the lineal relationship between molecularly distinct progenitor populations in the developing mouse hindbrain and mature norepinephrine neuron subtype identity. We have identified four genetically separable subpopulations of mature norepinephrine neurons differing in their anatomical location, axon morphology and efferent projection pattern. One of the subpopulations showed an unexpected projection to the prefrontal cortex, challenging the long-held belief that the locus coeruleus is the sole source of norepinephrine projections to the cortex. These findings reveal the embryonic origins of central norepinephrine neurons and provide multiple molecular points of entry for future study of individual norepinephrine circuits in complex behavioral and physiological processes including arousal, attention, mood, memory, appetite and homeostasis."*

### (Baral et al., 2022)

Spike-Dependent Dynamic Partitioning of the Locus Coeruleus Network through Noradrenergic Volume Release in a Simulation of the Nucleus Core

*"The Locus coeruleus (LC) modulates various neuronal circuits throughout the brain. Its unique architectural organization encompasses a net of axonal innervation that spans the entire brain; while its somatic core is highly compact"*

*".As the distances between noradrenergic neurons in the core of the LC are unusually small, neighboring neurons could theoretically impact each other via volume transmission of NE. We therefore set out to investigate if such interaction could be mediated through noradrenergic alpha2-receptors in a spiking neuron model of the LC"*

*".We determined the exact position of hundreds of LC neurons in the mouse brain stem via a tissue clearing approach and, based on this, further determined that 25 percent of noradrenergic neurons have a neighboring LC neuron within less than a 25-micrometer radius. By modeling NE diffusion, we estimate that more than 15 percent of alpha2-adrenergic receptors fraction can bind NE within such diffusion radius."*

### (Alam et al., 2023)

Electrophysiology as a Tool to Decipher the Network Mechanism of Visceral Pain in Functional Gastrointestinal Disorders

*"The brain nucleus locus coeruleus (LC) is located in the dorsolateral pons and is a significant source of norepinephrine (NE) in the brain (Figure 2). Retrograde tracing studies in rats demonstrated that ascending axons of noradrenergic LC neurons project to specific regions of the CNS including the thalamus, mPFC, ACC, hippocampus, hypothalamus, amygdala, and cerebellum (Berridge et al., 2003)(Mason et al., 1979), while descending axons project to the spinal dorsal horn (SDH) (Proudfit et al., 1991). The LC releases NE to modulate nociception by altering the excitability of spinothalamic tract neurons [131](Treviño et al., 1972)(Willis et al., 1974). These data suggest that LC and sub-coeruleus (SC) regions provide noradrenergic innervation to the forebrain, cerebellum, brainstem, and dorsal and ventral horns of the spinal cord (Proudfit et al., 1991)(Clark et al., 1991)(Sluka et al., 1992). Electrical stimulation of LC neurons produces antinociception (Janss et al., 1987)(Jones et al., 1986)(Jones et al., 1986)"*

*"A recent study used chemogenetic, the Designer Receptors Exclusively Activated by Designer Drugs (DREADD), to activate the LC-SC pathway selectively and explore its effect on neuropathic pain in mice. Chemogenetic activation of the LC-SC pathway increased the release of NE in the SDH and reduced pain intensity in neuropathic mice (Li et al., 2022)."*

### (Holland et al., 2021)

The role of noradrenaline in cognition and cognitive disorders

*"The forebrain noradrenergic input is from a small bilateral collection of neurons called the locus coeruleus (LC), where the cells begin rostrally at the level of the inferior colliculus adjacent to the cerebral aqueduct and end caudally near the lateral wall of the fourth ventricle (German et al., 1988) ; on axial brain slices, the anatomical landmarks are 1 mm under the fourth ventricle, 3 mm*

*from the midline, and centred 14-21 mm above the ponto-medullary junction (Fernandes et al., 2012) (Fig. 1)"*

*".Other conventional ways to identify noradrenaline producing neurons in the LC is by immunohistochemistry directed at tyrosine hydroxylase (TH), the enzyme that converts L-tyrosine to L-DOPA, or against dopamine betahydroxylase, which converts dopamine to noradrenaline. (Iversen et al., 1983) It is widely considered that the two markers are expressed by identical neuronal populations (Chan-Palay et al., 1989) representing the noradrenergic neuronal population of LC which constitutes more than 95% of neurons in the LC in controls. (Baker et al., 2004)(Oh et al., 2019)"*

*"Within the LC there is heterogeneity in both the population of the residing medium-sized neurons and neuronal numbers across the rostro-caudal gradient. The majority of medium-sized LC noradrenergic neurons are large multipolar cells (35-45 mm in diameter), with plump cell bodies and short dendrites; in the caudal LC and subcoeruleus, where the density of medium sized neurons is lower, the larger medium sized neurons are interspersed with smaller fusiform noradrenergic neurons (15 mm in diameter) with triangular cell bodies and two tufts of long dendrites. 19"*

## Cellular structure and morphological/electrophysiological features

*Mouse LC noradrenergic neurons are medium-sized cells with compact somata and dendrites that extend into peri-LC space, where nearby cells can interact. Physiologically, they are classic slow pacemakers that usually fire spontaneously at low rates, can switch into phasic burst-like responses, and are linked by dendritic gap junction coupling. (12 sources)*

At the cellular level, LC noradrenergic neurons are packed into a compact core, but their dendrites extend beyond the soma-rich center into surrounding peri-LC regions, which helps create local interactions despite the small size of the nucleus. Neighbor spacing is very tight: about 25% of mouse LC noradrenergic neurons have another LC neuron within 25 µm, making dendrite-to-dendrite and local transmitter-based interactions plausible. Recent multi-patch work further suggests that these neurons are not completely uniform, but can be grouped into at least two major morpho-electric types, with electrical coupling most often found between nearby neurons of the same type and arranged in small pairs, clusters, or chain-like local assemblies. Electrical coupling appears to occur mainly through dendrites rather than through conventional fast excitatory synapses between LC noradrenergic neurons. (Baral et al., 2022) (McKinney et al., 2023) (Gonye et al., 2023) (Ishimatsu et al., 1996)

Morphologically, the classic LC noradrenergic cell is a medium-to-large multipolar neuron with a plump soma and relatively short dendrites, although this varies across the rostrocaudal axis. A useful comparative description is that most LC noradrenergic neurons are large multipolar cells about 35–45 µm in diameter, while more caudal LC/subcoeruleus zones contain lower-density populations that also include smaller fusiform neurons around 15 µm with triangular cell bodies and two long dendritic tufts. More recent in vivo work in mouse also supports morphological variation within the LC core and surround, although TH-positive noradrenergic and nearby TH-negative cells can look broadly similar by simple morphology alone. (Holland et al., 2021) (Zouridis et al., 2024)

Electrophysiologically, LC noradrenergic neurons in mouse are identified by their spontaneous low-frequency pacemaker firing, usually around 1–2 Hz in acute slices and more broadly ~1–5 Hz across preparations. They show broad action potentials, a pronounced pacemaker-like baseline pattern, and in standard slice identification protocols they are recognized by tonic spontaneous firing together with absence of a clear hyperpolarization-activated current (Ih) and little or no membrane “sag” during hyperpolarizing current steps. In behaving-state frameworks, these neurons can also shift from tonic firing into phasic high-frequency discharge in response to salient stimuli, a key feature linked to state and attention control. (Drenan et al., 2008) (Wagner-Altendorf et al., 2019) (Gargano et al., 2020) (Kelly et al., 2023) (Gonye et al., 2023)

Although many basic firing properties are shared across LC neurons, there are measurable functional differences between projection-defined subgroups. In vitro comparison of mouse LC neurons projecting to medial prefrontal cortex, hippocampus, or cerebellum found broadly similar pacemaker and spike-generation properties, but different strengths of α2 autoreceptor-mediated inhibition: hippocampal-projecting neurons were more strongly inhibited by endogenous norepinephrine and by clonidine than prefrontal-projecting neurons. This suggests that some of the functional diversity of LC output is built into the intrinsic or local inhibitory physiology of different neurons, not just their axon targets. (Wagner-Altendorf et al., 2019)

LC neurons also receive strong excitatory input onto both somata and dendrites. In adult mouse, GluA1-containing AMPA receptor puncta are found on somata together with VGLUT2-positive inputs and on distal dendrites with VGLUT1-positive inputs, while GluA4 is seen mainly on distal dendrites; this supports compartment-specific glutamatergic drive onto LC neurons. In addition, many LC neurons are intrinsically sensitive to CO2/H+, which has been linked to modulation of respiratory control, especially through effects on potassium and calcium conductances. (Kelly et al., 2023) (Carvalho et al., 2014) (Gargaglioni et al., 2010)

Overall, the mouse LC is best viewed as a dense cluster of mostly noradrenergic pacemaker neurons with medium-sized somata, locally extending dendrites, and substantial electrotonic coupling. That shared scaffold gives the LC a coherent baseline rhythm, while differences in morphology, local coupling, receptor-mediated inhibition, and projection identity provide room for subpopulation-specific signaling. (Holland et al., 2021) (McKinney et al., 2023) (Wagner-Altendorf et al., 2019)

## Evidence

**(Baral et al., 2022)**

Spike-Dependent Dynamic Partitioning of the Locus Coeruleus Network through Noradrenergic Volume Release in a Simulation of the Nucleus Core

> *"The Locus coeruleus (LC) modulates various neuronal circuits throughout the brain. Its unique architectural organization encompasses a net of axonal innervation that spans the entire brain; while its somatic core is highly compact"*
>
> *".As the distances between noradrenergic neurons in the core of the LC are unusually small, neighboring neurons could theoretically impact each other via volume transmission of NE. We therefore set out to investigate if such interaction could be mediated through noradrenergic alpha2-receptors in a spiking neuron model of the LC"*
>
> *".We determined the exact position of hundreds of LC neurons in the mouse brain stem via a tissue clearing approach and, based on this, further determined that 25 percent of noradrenergic neurons have a neighboring LC neuron within less than a 25-micrometer radius. By modeling NE diffusion, we estimate that more than 15 percent of alpha2-adrenergic receptors fraction can bind NE within such diffusion radius."*

**(McKinney et al., 2023)**

Cellular composition and circuit organization of the locus coeruleus of adult mice

> *"The locus coeruleus (LC) is a small pontine nucleus that houses the vast majority of norepinephrine (NE)-producing neurons in the brain and sends widespread noradrenergic projections to all major divisions of the central nervous system (CNS) (Schwarz et al., 2015). By releasing NE across the CNS, the LC participates in many cognitive processes and behaviors crucial for survival, including fight and flight response, attentional control, and sleep/wake cycles"*
>
> *"LC noradrenergic neurons are diverse and could be classified into two major morpho-electric types. While fast excitatory synaptic transmission among LC noradrenergic neurons was not observed in our preparation, these mature LC neurons connected via gap junction at a rate similar to their early developmental stage and comparable to other brain regions. Most electrical connections form between dendrites and are restricted to narrowly spaced pairs or small clusters of neurons of the same type. In addition, more than two electrically coupled cell pairs were often identified across a cohort of neurons from individual multi-cell recording sets that followed a chain-like organizational pattern. The assembly of LC noradrenergic neurons thus follows a spatial and cell type-specific wiring principle that may be imposed by a unique chain-like rule."*

**(Gonye et al., 2023)**

Criteria for central respiratory chemoreceptors: experimental evidence supporting current candidate cell groups

> *"The locus coeruleus (LC) is a brainstem structure located in the rostral pons, lateral and ventral to the fourth ventricle; it comprises ~3000 noradrenergic neurons in mouse or rat, providing the primary noradrenergic innervation throughout the central nervous system (Loizou, 1969)Schwarz and Luo, 2015;(Liu et al., 2020)(McKinney et al., 2023). Its activity is tightly correlated to arousal levels and stress. Neurons within the LC are electrically coupled via gap junctions, particularly in their dendritic processes, and they exhibit a steady pacemaker-like baseline firing pattern with a pronounced subthreshold oscillation (Ishimatsu et al., 1996)(Oyamada et al., 1999)(Ballantyne et al., 2004)"*
>
> *".There is considerable evidence that LC neurons can influence respiration and display intrinsic CO 2 /H + sensitivity"*

**(Ishimatsu et al., 1996)**

Synchronous Activity in Locus Coeruleus Results from Dendritic Interactions in Pericoerulear Regions

*"Locus coeruleus (LC) neurons in brain slices from adult rats were studied using intracellular and extracellular recordings to investigate synchronous activity. Spontaneous field potentials were recorded with extracellular electrodes in solutions containing tetraethylammonium chloride (10 mm) and BaCl2 (1 mm). These field potentials were found throughout but not outside the LC cell body region. No field potentials were observed in control solutions. Paired recordings showed that field potentials were synchronous in all areas of the LC. The synchronous activity was resistant to tetrodotoxin (1 μm) and to the neurotransmitter receptor blockers d-2-amino-5-phosphonopentanoic acid, bicuculline, 6-cyano-7-nitroquinoxaline-2,3-dione, idazoxan, and strychnine, suggesting that this activity was not synaptically driven. Field potentials were also synchronous with oscillations in membrane potential recorded with intracellular electrodes. The oscillations in membrane potential were 5–30 mV in amplitude and had a biphasic waveform. Neither the frequency nor the waveform of the oscillations was dependent on the membrane potential. The glycyrrhetinic acid derivative carbenoxolone and intracellular acidification with CO2 disrupted synchronous activity, suggesting a role of electrotonic coupling. When the cell body region of the LC was isolated from the pericoerulear dendritic regions by sectioning the slice rostral and caudal to the cell body region, synchronous activity was reduced or abolished. Dendritic interaction in the pericoerulear region was also indicated by improved voltage control of the opioid-induced potassium current, as indicated by a shift in the reversal potential to the potassium equilibrium potential. The results suggest that electrical interactions between dendrites outside the cell body region can account for synchronous activity within the nucleus."*

**(Holland et al., 2021)**

The role of noradrenaline in cognition and cognitive disorders

*"The forebrain noradrenergic input is from a small bilateral collection of neurons called the locus coeruleus (LC), where the cells begin rostrally at the level of the inferior colliculus adjacent to the cerebral aqueduct and end caudally near the lateral wall of the fourth ventricle (German et al., 1988) ; on axial brain slices, the anatomical landmarks are 1 mm under the fourth ventricle, 3 mm from the midline, and centred 14-21 mm above the ponto-medullary junction (Fernandes et al., 2012) (Fig. 1)"*

*".Other conventional ways to identify noradrenaline producing neurons in the LC is by immunohistochemistry directed at tyrosine hydroxylase (TH), the enzyme that converts L-tyrosine to L-DOPA, or against dopamine betahydroxylase, which converts dopamine to noradrenaline. (Iversen et al., 1983) It is widely considered that the two markers are expressed by identical neuronal populations (Chan-Palay et al., 1989) representing the noradrenergic neuronal population of LC which constitutes more than 95% of neurons in the LC in controls. (Baker et al., 2004)(Oh et al., 2019)"*

*"Within the LC there is heterogeneity in both the population of the residing medium-sized neurons and neuronal numbers across the rostro-caudal gradient. The majority of medium-sized LC noradrenergic neurons are large multipolar cells (35-45 mm in diameter), with plump cell bodies and short dendrites; in the caudal LC and subcoeruleus, where the density of medium sized neurons is lower, the larger medium sized neurons are interspersed with smaller fusiform noradrenergic neurons (15 mm in diameter) with triangular cell bodies and two tufts of long dendrites. 19"*

**(Zouridis et al., 2024)**

Juxtacellular recordings from identified neurons in the mouse locus coeruleus

*"The locus coeruleus (LC) is the primary source of noradrenergic transmission in the mammalian central nervous system. This small pontine nucleus consists of a densely packed nuclear core—which contains the highest density of noradrenergic neurons—embedded within a heterogeneous surround of non-noradrenergic cells"*

*". We found that noradrenergic neurons (identified by tyrosine hydroxylase, TH, expression; TH-positive) and intermingled putatively non-noradrenergic (TH-negative) cells displayed similar morphologies and responded to foot shock stimuli with excitatory responses; however, on average, TH-positive neurons exhibited more prominent foot shock responses and post-activation firing suppression. The two cell classes also displayed different spontaneous firing rates, spike waveforms and temporal spiking properties."*

**(Drenan et al., 2008)**

In Vivo Activation of Midbrain Dopamine Neurons via Sensitized, High-Affinity α6* Nicotinic Acetylcholine Receptors

*"(A) Diagram of coronal sections (bregma À5.4 mm) containing locus coeruleus (Franklin and Paxinos, 2008). (B) Tyrosine hydroxylase (TH) staining of NE neurons in coronal slices. Coronal slices (200 mm; bregma À5.4 mm) from an a6 L9 0 S mouse were fixed and stained for TH as described in Supplemental Experimental Procedures. The boxed area from (A) is shown. Scale bar: 100 mm. MPB, medial parabrachial nucleus; LC, locus coeruleus; Bar, Barrington's nucleus; LDTg, laterodorsal tegmental nucleus; 4V, fouth ventricle. (C) Electrophysiological identification of LC neurons. LC neurons are identified by (Ci) pacemaker firing at 1-2 Hz, (Cii) lack of I h , and (Ciii) lack of membrane potential "sag" for hyperpolarizing current pulses (responses shown for injection of À80, À40, and +20 pA)"*

*".(D) Hypersensitive nicotinic responses in LC neurons. Patch-clamp recordings were taken from LC neurons in coronal slices from WT, a4 L9 0 A , and a6 L9 0 S mice. Neurons were voltage clamped at À60 mV, and nicotine at the indicated concentration was locally applied (arrow) to the cell body (250 ms)."*

**(Wagner-Altendorf et al., 2019)**

Axonal projection-specific differences in somatodendritic α2 autoreceptor function in locus coeruleus neurons

*"The pontine nucleus locus coeruleus (LC) contains the majority-about 10,000-15,000 neurons on each side in humans and 1,500 in rodents-of central noradrenergic neurons that project to a wide variety of targets including cerebellum, prefrontal cortex and hippocampus (Berridge et al., 2003)(Moore et al., 1979)(Swanson, 1976)"*

*".Electrophysiological studies of LC neurons have identified several characteristic functional features such as low-frequency pacemaker activity with broad action potentials, transient high-frequency burst discharges in response to salient stimuli and an apparently homogeneous inhibition of firing by activation of somatodendritic α2 autoreceptors (α2AR)"*

*".Using fluorescent beads-mediated retrograde tracing in adult C57Bl6/N mice, we compared the anatomical distributions and functional in vitro properties of identified LC neurons projecting either to medial prefrontal cortex, hippocampus or cerebellum. The functional in vitro analysis of LC neurons confirmed their mostly uniform functional properties regarding action potential generation and pacemaker firing. However, we identified significant differences in tonic and evoked α2AR-mediated responses. While hippocampal-projecting LC neurons were partially inhibited by endogenous levels of norepinephrine and almost completely silenced by application of saturating concentrations of the α2 agonist clonidine, prefrontal-projecting LC neurons were not affected by endogenous levels of norepinephrine and only partially inhibited by saturating concentrations of clonidine."*

**(Gargano et al., 2020)**

Lack of Cannabinoid Receptor Type-1 Leads to Enhanced Age-Related Neuronal Loss in the Locus Coeruleus

*"The locus coeruleus (LC), or "blue spot", is a small nucleus located in the pons of the brainstem, lateral to the IVth ventricle. It projects throughout the neuroaxis and represents the major source of noradrenaline (NE) in the central nervous system. Although the LC consists of a relatively low number of noradrenergic neurons (45-50 thousand in a normal healthy young adult human (Sharma et al., 2010) and approximately 1500 in adult mice (Sturrock et al., 1985)), its projecting area is extremely wide; noradrenergic terminals are abundantly present in the cortex, mainly in the somatosensory and motor cortex (Gaspar et al., 1989), hippocampus and amygdala (Gaspar et al., 1989)(Sadikot et al., 1990)(Radley et al., 2008), hypothalamus (Ginsberg et al., 1993) and in the brainstem itself (Westlund et al., 1980). The LC also receives a large number of afferents, mainly from the cortex, amygdala [8] (Pammer et al., 1990) and from the spinal cord (Westlund et al., 1996)"*

*".Electrophysiological studies revealed that LC noradrenergic neurons have a peculiar double firing mode; they are able to switch between tonic or phasic mode, regulating different behavioral states. A tonic firing is related to a "low attention" mode, while the switch to the phasic firing happens in response to a relevant "focus-demanding" stimuli (Aston-Jones et al., 2005). These studies suggested that LC may play a role in modulating numerous cognitive functions, including attention, memory as well as in the regulation of sleep-wake states and stress response (Benarroch, 2017). Under physiological conditions, LC neurons respond to external and internal sensory stimuli and this response influences learning and memory processes (Uematsu et al., 2015)."*

**(Kelly et al., 2023)**

Subunit-specific expression and function of AMPA receptors in the mouse locus coeruleus

*"The locus coeruleus (LC) is a collection of neurons, located bilaterally within the dorsal pons, adjacent to the fourth ventricle (Dahlstroem et al., 1964). LC neurons are distinguished from surrounding structures by their production of the neurotransmitter noradrenaline (NA), which in turn facilitates the brain-wide modulatory role of this nucleus (Berridge & Waterhouse, 2003). By projecting to almost the entire brain, the LC-NA system modulates ongoing brain activity, thereby influencing some of the most salient brain functions, including cognition (Usher et al., 1999), memory (Sara, 2015), arousal (Carter et al., 2010)), mood (McCall et al., 2015) and the central limb of the stress response (Valentino et al., 2008). In addition, the LC also modulates the physiology of various peripheral systems including the control of respiration (Carvalho et al., 2014)(Gargaglioni et al., 2010) and cardiac function (Biancardi et al., 2014)(Patrone et al., 2014)(Yao et al., 2005)"*

*".Central to LC function, in both health and disease, is the spontaneous firing rate (FR) of the principal neurons which determine the pattern and amount of NA release"*

*".In the awake state and in in vitro preparations, LC neurons characteristically fire action potentials spontaneously, with a frequency ranging from ~1 to 5 Hz (Berridge & Waterhouse, 2003)"*

*".GluA1 immunoreactive clusters were associated with puncta immunoreactive for VGLUT2 on somata, and VGLUT1 on distal dendrites. GluA4 was associated with these synaptic markers only in the distal dendrites. No specific signal was detected for the GluA2-3 subunits."*

**(Carvalho et al., 2014)**

Neurochemical and electrical modulation of the locus coeruleus: contribution to CO2drive to breathe

*"The locus coeruleus (LC) is a dorsal pontine region, situated bilaterally on the floor of the fourth ventricle. It is considered to be the major source of noradrenergic innervation in the brain"*

*". The LC region is a densely packed nucleus, comprising about 1500 neurons per side (Aston- (Aston-Jones et al., 1984)"*

*". In addition to NA, the cell bodies also secret several neuropeptides (Sutin et al., 1991), among which are vasopressin, somatostatin, neuropeptide Y, enkephalin, neurotensin, CRH, and galanin (for review, see (Olpe et al., 1991)"*

*". LC NA neurons projects broadly throughout the neuraxis, from spinal cord to neocortex (Swanson and Hartman, 1975;Foote et al., 1983;Berridge and Waterhouse, 2003). In fact, it is estimated that  50%of all the noradrenergic projections in the central nervous system (CNS) originate in the LC which are directed toward the forebrain, cerebellum, brainstem and spinal cord (Aston- Jones et al., 1995;Berridge and Waterhouse, 2003). LC is the primary source of an extensive, yet regionally specialized, noradrenergic innervation of the forebrain (Berridge and Waterhouse, 2003) and it is considered a major wakefulness promoting nucleus with activation of the LC resulting in an increase in EEG signs of alertness (Samuels and Szabadi, 2008)."*

**(Gargaglioni et al., 2010)**

THE LOCUS COERULEUS AND CENTRAL CHEMOSENSITIVITY

*"The locus coeruleus (LC) lies in the dorsal pons and supplies noradrenergic (NA) input to many regions of the brain, including respiratory control areas. The LC may provide tonic input for basal respiratory drive and is involved in central chemosensitivity since focal acidosis of the region stimulates ventilation and ablation reduces CO2-induced increased ventilation. The output of LC is modulated by both serotonergic and glutamatergic inputs. A large percentage of LC neurons are intrinsically activated by hypercapnia. This percentage and the magnitude of their response are highest in young neonates and decrease dramatically after postnatal day P10. The cellular bases for intrinsic chemosensitivity of LC neurons are comprised of multiple factors, primary among them being reduced extracellular and intracellular pH, which inhibit inwardly rectifying and voltage-gated K+ channels, and activate L-type Ca2+ channels. Activation of KCa channels in LC neurons may limit their ultimate response to hypercapnia. Finally, the LC mediates central chemosensitivity and contains pH-sensitive neurons in amphibians, suggesting that the LC has a long-standing phylogenetic role in respiratory control."*

## Molecular markers and neurochemical identity

*Mouse LC neurons are defined mainly by their noradrenergic identity. In practice, the most used markers are TH and DBH, while transcriptomic and recent molecular studies add transporter/synthesis genes, developmental transcription factors, and co-expressed neuropeptides such as galanin. (11 sources)*

- **Core neurochemical identity: noradrenergic neurons**
- LC neurons are the main central source of **norepinephrine/noradrenaline**, and the LC contains the largest and densest cluster of these neurons in the mouse CNS. (Liu et al., 2013) (Lutsenko et al., 2019) (Lee et al., 2024)
- In the adult LC, the great majority of neurons are noradrenergic, commonly described as **>95% of LC neurons** in control tissue. (Kaalund et al., 2020) (Holland et al., 2021)
- **Most used histological markers**
- **Tyrosine hydroxylase (TH)** is the standard marker used to identify mouse LC noradrenergic neurons. TH is the rate-limiting enzyme in catecholamine synthesis and is routinely used for LC localization and cell counting. (Liu et al., 2013) (Loucano et al., 2022) (Lee et al., 2024)
- **Dopamine beta-hydroxylase (DBH)** is another key marker. Because DBH converts dopamine to noradrenaline, it is often used alongside TH and is especially useful as a direct marker of noradrenergic phenotype. (Lutsenko et al., 2019) (Kaalund et al., 2020) (Lee et al., 2024)
- TH and DBH are widely treated as marking the **same LC noradrenergic population**. (Kaalund et al., 2020) (Holland et al., 2021)
- For practical mouse work, **DBH or TH immunostaining** is commonly used to find the tiny LC in sections. (Lutsenko et al., 2019) (Schmidt et al., 2019)
- **Important caveat on TH labeling**
- TH is highly useful, but it is not perfectly one-to-one with all LC noradrenergic states. Some reports note **Phox2a-positive / TH-negative "resting noradrenergic" neurons**, especially in rostral LC, suggesting that noradrenergic identity can persist even when TH is not strongly expressed at baseline. (Terrier et al., 2024)

- Dense TH-positive neuropil in the LC can also make individual somata harder to resolve in standard immunostains. (Loucano et al., 2022)
- **Developmental / identity transcription factors**
- **Phox2a** can mark LC noradrenergic neurons and has been used experimentally as a noradrenergic-cell marker in the mouse LC. (Terrier et al., 2024)
- More broadly, Phox2-family transcription factors are well-established parts of the developmental program of brainstem noradrenergic neurons. (Terrier et al., 2024)
- **Molecular markers from profiling studies**
- A notable mouse LC molecular profiling study found strong enrichment of genes needed for norepinephrine synthesis, packaging, and reuptake, including **Th, Dbh, Slc6a2 (NET), Slc18a2 (VMAT2), and Gch1**; these are useful molecular markers of LC noradrenergic neurons. (Mulvey et al., 2018)
- These markers fit the canonical noradrenergic pathway: catecholamine synthesis from tyrosine through dopamine to norepinephrine, vesicular loading, and reuptake through the **NET transporter**. (Slavova et al., 2024) (Mulvey et al., 2018)
- **Receptors and transmitter machinery**
- LC norepinephrine neurons act through **adrenergic receptors**, with **α2 autoreceptors** being especially important for self-inhibition, while released NE acts on α1 and β receptor families in targets. (Slavova et al., 2024)
- NE clearance and breakdown involve **NET/Slc6a2**, with degradation by **MAO-A** and **COMT**. (Slavova et al., 2024)
- **Co-transmitters / co-expressed neuropeptides**
- Mouse LC NE neurons are not purely monoaminergic. Single-cell RNA-seq-based work found that they express **galanin** and transcripts for **more than 19 other neuropeptides** as well as **more than 30 neuropeptide receptors**, including **Galr1** at low levels. (Caramia et al., 2023)
- Review-level summaries also describe LC NE cells as co-releasing **galanin**, and in some accounts **CRF** and **NPY** are discussed in the LC region; however, at least one recent mouse-focused source states that **NPY in mice is exclusive to peri-LC GABAergic cells rather than NE neurons**, so NPY should be treated as a peri-LC marker, not a firm marker of LC noradrenergic neurons themselves. (Slavova et al., 2024) (Caramia et al., 2023)
- **Relation to nearby non-noradrenergic cells**
- The broader LC region includes a **peri-LC GABAergic population** around the noradrenergic core, so marker choice matters when distinguishing true LC NE neurons from surrounding local circuit neurons. (Slavova et al., 2024)
- **A practical marker set for mouse LC noradrenergic neurons**
- For most mouse studies, the most reliable marker combination is **TH+ / DBH+**, optionally supported by **Slc6a2/NET**, **Slc18a2/VMAT2**, and developmental identity markers such as **Phox2a**. (Kaalund et al., 2020) (Holland et al., 2021) (Mulvey et al., 2018) (Terrier et al., 2024)

## Evidence

(Liu et al., 2013)

Degenerative alterations in noradrenergic neurons of the locus coeruleus in Alzheimer's disease

*"The locus coeruleus is located in the posterior area of the rostral pons in the lateral floor of the fourth ventricle. It is the major site to secrete the catecholamine neurotransmitter, noradrenaline, and is therefore also known as a noradrenergic cell group (Bouret et al., 2005) . The locus coeruleus has over 50% of the noradrenergic neurons in the entire central nervous system, and is the principal site for brain synthesis of noradrenaline. The nucleus has widespread projections that innervate the spinal cord, cerebellum, hypothalamus, thalamic relay nuclei, amygdala, basal telencephalon, and cortex"*

*". Noradrenaline can regulate the regional microenvironment, and functions both as a neurotransmitter and as an endogenous anti-inflammatory factor (Feinstein et al., 2002)(Heneka et al., 2006)"*

*". Noradrenergic neurons can be visualized by tyrosine hydroxylase, a marker of catecholaminergic neurons"*

*". Immunohistochemistry for tyrosine hydroxylase (a marker of catecholaminergic neurons in the locus coeruleus) revealed that APP/PS1 mice had 23% fewer cells in the locus coeruleus compared with aged wild-type mice. APP/PS1 mice also had increased numbers of cell bodies of neurons positive for tyrosine hydroxylase, but fewer tyrosine hydroxylase-positive fibers, which were also short, thick and broken. Quantitative analysis using unbiased stereology showed a significant age-related increase in the mean volume of tyrosine droxylase-positive neurons in aged APP/PS1 mice compared with young APP/PS1 mice. Moreover, the mean volume of tyrosine hydroxylase-positive neurons was positively correlated with the total volume of the locus coeruleus"*

*". We studied the morphology of noradrenergic neurons in the locus coeruleus using tyrosine hydroxylase immunohistochemistry as a specific marker of catecholaminergic neurons, and quantitatively analyzed changes in the number of tyrosine hydroxylase-positive neurons and fibers using unbiased stereology."*

**(Lutsenko et al., 2019)**

COPPER AND THE BRAIN NORADRENERGIC SYSTEM

*"Locus coeruleus (LC) represents a cluster of approximately 1500 neurons located in the pons (pontine tegmentum) region near the 4th ventricle (Fig. 4, upper panel) (Grimm et al., 2004)"*

*".The noradrenergic neurons are concentrated in the LC. Although LC is small, it sends projections throughout the brain and represents the major source of norepinephrine in the CNS. The LC neurons are responsible for modulation of arousal, attention, and vigilance; they accomplish this function through regulation of target regions in a forebrain (Chamberlain et al., 2013)(Sara, 2009) (Szabadi, 2013). In the coronal sections of the brain (Fig. 4, lower panel), LC has a characteristic "horn-like" appearance at the edges of the ventricle and can be easily identified by immunostaining for the Cu-dependent enzyme dopamine-β-hydroxylase, DBH, or tyrosine hydroxylase, which is an established marker of noradrenergic neurons (Fig. 4, lower panel)."*

**(Lee et al., 2024)**

Sex Differences in Single Neuron Function and Proteomics Profiles Examined by Patch-Clamp and Mass Spectrometry in the Locus Coeruleus of the Adult Mouse

*"A1 noradrenergic neurons in the medulla, A2 neurons in the nucleus tractus solitarius and A6 neurons in the LC project to many areas in the CNS, including the cortex, hippocampus, thalamus, and spinal cord 18 . The LC, a small nucleus located on either side of the fourth ventricle, contains the largest cluster of noradrenergic neurons in the CNS. LC neurons project to the cortex, hippocampus, amygdala, hypothalamus, cerebellum, and spinal cord 18 and modulate physiological processes, such as arousal, attention, cognition, anxiety, and responses to stress 18-21"*

*"We identified LC noradrenergic neurons in the pons by their location along the lateral margin of the fourth ventricle 28, 29 and by expression of tyrosine hydroxylase (TH) 26, a key enzyme involved in catecholamine synthesis. We also labeled LC neurons with anti-dopamine beta-hydroxylase (DBH) antibodies, an enzyme that converts dopamine to noradrenaline and is a marker of noradrenergic neurons 26, 30."*

**(Kaalund et al., 2020)**

Locus coeruleus pathology in progressive supranuclear palsy, and its relation to disease severity

*"The locus coeruleus is the major source of noradrenaline to the brain and contributes to a wide range of physiological and cognitive functions including arousal, attention, autonomic control, and adaptive behaviour"*

*". Another way to identify the locus coeruleus cells producing noradrenaline is via the immuno-staining for tyrosine hydroxylase, the enzyme that converts L-tyrosine to L-DOPA before the conversion of L-DOPA to noradrenaline. Immunohistochemistry against the dopamine betahydroxylase, the enzyme that transform dopamine into noradrenaline, can also be used. These two biochemical markers are expressed by the same neuronal population [15] i.e., the noradrenergic cells that constitutes more than 95% of neurons in the locus coeruleus [2,39]. However, a proportion of tyrosine hydroxylase positive neurons, which are more numerous in the rostral than caudal pons, lack pigmentation [2,15,25]. The number of such neurons is limited in relation to the total neuronal number in the locus coeruleus [2,15]25]."*

(Holland et al., 2021)

The role of noradrenaline in cognition and cognitive disorders

*"The forebrain noradrenergic input is from a small bilateral collection of neurons called the locus coeruleus (LC), where the cells begin rostrally at the level of the inferior colliculus adjacent to the cerebral aqueduct and end caudally near the lateral wall of the fourth ventricle (German et al., 1988) ; on axial brain slices, the anatomical landmarks are 1 mm under the fourth ventricle, 3 mm from the midline, and centred 14-21 mm above the ponto-medullary junction (Fernandes et al., 2012) (Fig. 1)"*

*".Other conventional ways to identify noradrenaline producing neurons in the LC is by immunohistochemistry directed at tyrosine hydroxylase (TH), the enzyme that converts L-tyrosine to L-DOPA, or against dopamine betahydroxylase, which converts dopamine to noradrenaline. (Iversen et al., 1983) It is widely considered that the two markers are expressed by identical neuronal populations (Chan-Palay et al., 1989) representing the noradrenergic neuronal population of LC which constitutes more than 95% of neurons in the LC in controls. (Baker et al., 2004)(Oh et al., 2019)"*

*"Within the LC there is heterogeneity in both the population of the residing medium-sized neurons and neuronal numbers across the rostro-caudal gradient. The majority of medium-sized LC noradrenergic neurons are large multipolar cells (35-45 mm in diameter), with plump cell bodies and short dendrites; in the caudal LC and subcoeruleus, where the density of medium sized neurons is lower, the larger medium sized neurons are interspersed with smaller fusiform noradrenergic neurons (15 mm in diameter) with triangular cell bodies and two tufts of long dendrites. 19"*

(Loucano et al., 2022)

Pain Modulation from the Locus Coeruleus in a Model of Hydrocephalus: Searching for Oxidative Stress-Induced Noradrenergic Neuroprotection

*"Tyrosine hydroxylase (TH), the enzyme catalyzing the rate-limiting step in catecholamine biosynthesis, is frequently used in the immunohistochemical identification of noradrenergic LC neurons (Bucci et al., 2017)(Arikawe et al., 2021) and is even considered a noradrenergic marker of the LC (Hoffmeister et al., 2021)"*

*"TH-immunoreaction at the LC was recognized by dense brown immunostaining (Figure 2A,B). The intense staining precluded the identification of neurons immunoreactive to TH (TH-IR) which was mostly due to the density of the neuropil, as described previously (Grzanna et al., 1980)[34]"*

*"The images in Figure 2A,B depict approximately the same rostrocaudal level (around 0,8 mm caudal to the interaural line)."*

(Schmidt et al., 2019)

Localization of the Locus Coeruleus in the Mouse Brain

*"The locus coeruleus (LC) is an important region in the brainstem and a major site of norepinephrine (NE) production (Robertson et al., 2013) . The LC sends projections throughout the brain (Kobayashi et al., 1975) to the cortex, the hippocampus and the cerebellum (Olson et al., 1971) and regulates major physiological processes, including circadian rhythm (Costa et al., 2018)(Semba et al., 1984) , attention and memory (Takeuchi et al., 2016) , stress (Korf et al., 1973) , cognitive processes (Sara et al., 1991) , and emotion 9,(File et al., 1979)"*

*"Due to their small size, the mouse LC has an average diameter of ~300 µm, leading to difficulty locating the structure"*

*"Here, we describe a step-by-step immunohistochemistry-based protocol to localize the LC in the mouse brain. Dopamine-β-hydroxylase (DBH), and alternatively, tyrosine hydroxylase (TH), both enzymes highly expressed in the LC, are used as immunohistochemical markers in brain slices."*

(Terrier et al., 2024)

Long-term olfactory enrichment promotes non-olfactory cognition, noradrenergic plasticity and remodeling of brain functional connectivity in older mice.

*"Using immunohistochemistry of the transcription factor Phox2a specifically expressed in noradrenergic neurons (Goridis et al., 1999)(Morin et al., 1997)(Yang et al., 1998) (Fig. 5A), we measured the surface of the Locus Coeruleus (data not shown) and the density of noradrenergic neurons (Fig. 5B)"*

*".Similar densities of Tyrosine Hydroxylase positive cells (TH, enzyme required for NA synthesis and marker of NA neurons) were also found in the Locus Coeruleus of enriched and control mice (Fig. 5A and C, see Table 5 for detailed statistical results). Moreover, "resting-noradrenergic" neurons exist in the rostral Locus Coeruleus under basal conditions. These neurons express the Phox2a transcription factor but not TH and can be reactivated following treatment with RU 24722 (a catecholaminergic activator) (Bezin et al., 2000)."*

(Mulvey et al., 2018)

Molecular and Functional Sex Differences of Noradrenergic Neurons in the Mouse Locus Coeruleus

*"The locus coeruleus (LC) is the largest source of forebrain norepinephrine (NE), and abnormalities in LC-NE function contribute to diseases such as anxiety, depression, post-traumatic stress disorder (PTSD), and attention deficit hyperactive disorder (ADHD)"*

*".In the mouse, the LC is comprised of approximately 2,000 neurons per side, the majority of which are uniquely noradrenergic (Robertson et al., 2013; Chandler et al., 2014)"*

*".We profiled mouse noradrenergic neurons in vivo, defining over 3,000 high-confidence transcripts expressed therein, including druggable receptors"*

*".To compare molecular differences in female and male LC neurons, we developed a translational profiling strategy by combining the bacTRAP technology with a tyrosine hydroxylase driver line, generating Th-EGFP-L10a transgenic mice"*

*".we observed GFP-positive cells in catecholaminergic populations throughout the brain and retina, including in the LC, ventral tegmental area (VTA), substantia nigra pars compacta (SNc), hypothalamic arcuate nucleus, and olfactory bulb, among others, and in retinal amacrine and ganglion cells"*

*".We additionally stained transgenic LC sections for DBH, and we observed 100% co-expression of the EGFP-L10a transgene in TH-positive, DBH-positive LC neurons (Figure 1A)"*

*".We then used this line to molecularly profile female and male LC neurons under baseline conditions"*

*".We generated four biological replicates each of female and male adult LC neuron input and immunoprecipitation (IP) samples"*

*".In total, we identified 3,045 genes significantly enriched in LC neurons, and these represent a molecular profile of LC neurons in the mouse"*

*".We note that genes required for NE synthesis and release, including Gch1, Slc6a2, Slc18a2, Dbh, and Th, are highly enriched in the LC translatome, and therefore constitute useful markers for this neuronal population."*

**(Slavova et al., 2024)**

Role of the locus coeruleus-noradrenergic system in stress-related psychopathology and resilience: clinical and pre-clinical evidences.

*"The locus coeruleus (LC) region is divided into two parts: the LC, which primarily contains noradrenergic cells (NE, depicted in blue), and the peri-LC, which includes GABAergic neurons (depicted in grey). Within the noradrenergic cells, NE is synthesized through the conversion of Tyrosine (Tyr) to DOPA (3,4dihydroxyphenylalanine), then to Dopamine (DA), and finally to NE, which is released into the synaptic cleft. NE primarily acts on four types of receptors: α1, β1, β2, and α2 (an auto receptor). The degradation of NE occurs via the enzymes COMT (postsynaptic) and MAO-A (presynaptic) after reuptake by the NET transporter"*

*".NE cells simultaneously secrete Galanin (GAL), Neuropeptide Y (NPY), and corticoliberine (CRF) acting on GAL1R, various NPY receptors (Y1, Y2, Y5), and CRF1R respectively. In mice, NPY is exclusively secreted by GABAergic cells in the peri-LC."*

**(Caramia et al., 2023)**

Neuronal diversity of neuropeptide signaling, including galanin, in the mouse locus coeruleus

*"The locus coeruleus (LC) is the main source of norepinephrine (NE) in the brain, modulates wakefulness, and is altered in stress-related disorders"*

*".Here, we combine single-cell-RNA-sequencing, neuroanatomy, and electrophysiology to build a map of galanin and its receptors in the mouse LC. We found that NE neurons express mRNA transcripts for galanin, >19 other neuropeptides (but Npy), and >30 neuropeptide receptors with Galr1 included at low levels. Selective activation of GalR1 inhibited the spontaneous activity of a subset of NE neurons possibly through the activation of Kv currents."*

# Functions and projection-defined heterogeneity

*LC noradrenergic neurons help set brain state and can affect arousal, attention, learning, stress, pain, memory, sleep, and some autonomic functions. Even though LC output is widespread, it is not fully uniform: different LC neurons project to different targets and can show different response patterns, receptor sensitivity, and behavioral roles. (20 sources)*

Functionally, mouse LC noradrenergic neurons act as a broad modulatory system that can rapidly change how other brain regions process information. Across studies, the LC is linked to arousal and wakefulness, attention and vigilance, learning and memory, stress and anxiety, pain control, and control of respiration and other autonomic functions. Its impact is so broad because LC axons reach much of the neuraxis, including cortex, hippocampus, amygdala, hypothalamus, cerebellum, brainstem, and spinal cord. (Carvalho et al.,

2014) (Lutsenko et al., 2019) (Cao et al., 2020) (Gargano et al., 2020) (Kelly et al., 2023) (Lee et al., 2024) (Zhou et al., 2024)

A classic systems view is that LC firing mode matters. Tonic activity is associated with general arousal state and lower-selectivity monitoring, whereas phasic responses are triggered by salient or task-relevant events and support focused responding, alertness, and learning-related signaling. In mice, causal work supports this idea: LC activity can promote wakefulness and arousal, and stress-related increases in tonic LC firing are sufficient to drive anxiety-like and aversive behavior. (Wagner-Altendorf et al., 2019) (Gargano et al., 2020) (Carter et al., 2010) (Cao et al., 2020) (McCall et al., 2015)

The LC is also important for learning and memory, but likely through partially specialized output channels rather than a single global signal. Review and tracing work argues that subpopulations of LC neurons innervate specific regions such as medial prefrontal cortex and amygdala, supporting different aspects of learning, including cognitive flexibility versus emotional associative learning. (Uematsu et al., 2015) (Gargano et al., 2020) In the hippocampal system, mouse studies further show that LC TH+ neuron activity is strongly engaged by novelty and can enhance memory retention, with evidence consistent with catecholamine-dependent plasticity in CA1. (Cao et al., 2020) (Takeuchi et al., 2016)

Projection-defined heterogeneity is now a key part of the mouse LC picture. Although many LC neurons share similar pacemaker and spike-generation properties, retrograde tracing and slice physiology show that neurons projecting to different targets are not identical. In one mouse study, medial prefrontal-, hippocampal-, and cerebellum-projecting LC neurons had broadly similar baseline electrophysiology, but differed in α2 autoreceptor-mediated inhibition: hippocampal-projecting cells were more strongly inhibited by endogenous norepinephrine and by clonidine than prefrontal-projecting cells. This means that target-defined LC neurons can differ in how strongly they self-regulate their own output. (Wagner-Altendorf et al., 2019)

Recent mouse work extends this idea from single targets to cortical modules. During learned behavior, transient LC-NA activity can carry at least two separable signals: one that precedes action and helps task execution, and another that follows reinforcement and helps improve later performance. These signals are not distributed the same way across outputs: behavioral execution signals were stronger in motor cortex projections, while negative reinforcement signals were more widely shared across cortical targets. (Breton-Provencher et al., 2022) Newer single-neuron reconstruction and activity studies further suggest that individual LC neurons often have broad but still selective arborization patterns, with projection bias and graded molecular identity related to soma position inside the LC; for example, dorsal LC neurons with cortical projections were linked to choice switching and reward-prediction-error-like signals, while ventral LC neurons showed higher background activity when mice ignored reward-predicting cues. (Su et al., 2026) (Su et al._1, 2026)

Functionally important heterogeneity also appears in descending and local circuits. LC projections to the spinal dorsal horn suppress nociceptive transmission, and selective activation of the LC spinal cord pathway in mice reduces neuropathic pain and spinal neuroinflammation through norepinephrine release and α2-adrenergic signaling. (Alam et al., 2023) (Li et al., 2022) The LC is also part of stress and affect circuits: amygdala-derived inputs, including GABAergic and CRH-related pathways, can shape LC activity, and this input organization is consistent with a direct route by which emotional circuits regulate LC output and anxiety-like states. (Dimitrov et al., 2013) (Cao et al., 2020) (McCall et al., 2015)

Beyond cortex and spinal cord, LC output is specialized across many other targets. Examples in the mouse include projections to dorsal raphe that can excite serotonin neurons through α1-adrenergic mechanisms while also being suppressed by presynaptic α2 autoreceptor action, and direct projections to olfactory structures such as anterior piriform cortex and olfactory bulb. These examples reinforce the idea that LC neurons do not simply broadcast one identical signal everywhere. (Gugel et al., 2024) (Geng et al., 2025)

Overall, the best current view is that the mouse LC combines **global reach with partial modularity**. It can coordinate whole-brain state, but it also contains subpopulations defined by projection target, receptor

physiology, anatomical position, and behavioral coding. This organization likely allows the same small noradrenergic nucleus to support both general state control and more specific functions such as cortical task engagement, reinforcement learning, stress responses, and analgesia. (Uematsu et al., 2015) (Breton-Provencher et al., 2022) (Su et al., 2026) (Su et al._1, 2026)

## Evidence

### (Carvalho et al., 2014)

Neurochemical and electrical modulation of the locus coeruleus: contribution to CO2drive to breathe

*"The locus coeruleus (LC) is a dorsal pontine region, situated bilaterally on the floor of the fourth ventricle. It is considered to be the major source of noradrenergic innervation in the brain"*

*". The LC region is a densely packed nucleus, comprising about 1500 neurons per side (Aston- (Aston-Jones et al., 1984)"*

*". In addition to NA, the cell bodies also secret several neuropeptides (Sutin et al., 1991), among which are vasopressin, somatostatin, neuropeptide Y, enkephalin, neurotensin, CRH, and galanin (for review, see (Olpe et al., 1991)"*

*". LC NA neurons projects broadly throughout the neuraxis, from spinal cord to neocortex (Swanson and Hartman, 1975;Foote et al., 1983;Berridge and Waterhouse, 2003). In fact, it is estimated that ≈ 50%of all the noradrenergic projections in the central nervous system (CNS) originate in the LC which are directed toward the forebrain, cerebellum, brainstem and spinal cord (Aston- Jones et al., 1995;Berridge and Waterhouse, 2003). LC is the primary source of an extensive, yet regionally specialized, noradrenergic innervation of the forebrain (Berridge and Waterhouse, 2003) and it is considered a major wakefulness promoting nucleus with activation of the LC resulting in an increase in EEG signs of alertness (Samuels and Szabadi, 2008)."*

### (Lutsenko et al., 2019)

COPPER AND THE BRAIN NORADRENERGIC SYSTEM

*"Locus coeruleus (LC) represents a cluster of approximately 1500 neurons located in the pons (pontine tegmentum) region near the 4th ventricle (Fig. 4, upper panel) (Grimm et al., 2004)"*

*".The noradrenergic neurons are concentrated in the LC. Although LC is small, it sends projections throughout the brain and represents the major source of norepinephrine in the CNS. The LC neurons are responsible for modulation of arousal, attention, and vigilance; they accomplish this function through regulation of target regions in a forebrain (Chamberlain et al., 2013)(Sara, 2009) (Szabadi, 2013). In the coronal sections of the brain (Fig. 4, lower panel), LC has a characteristic "horn-like" appearance at the edges of the ventricle and can be easily identified by immunostaining for the Cu-dependent enzyme dopamine-β-hydroxylase, DBH, or tyrosine hydroxylase, which is an established marker of noradrenergic neurons (Fig. 4, lower panel)."*

### (Cao et al., 2020)

Fast Localization and Sectioning of Mouse Locus Coeruleus

*"LC is a multifunctional nucleus which is also the source of norepinephrine in the brain"*

*". Locus coeruleus-(LC-) related researches are important parts of neuroscience because it is not only a key hub of noradrenergic neurons, but also the only (or most important) source of norepinephrine in the brain [1]. LC projects into the brain to participate in the regulation of circadian rhythm [2], stress [3], memory and cognition [4], general anesthesia [5], and so on. LC also innervates the spinal cord to regulate physiological and pathological pain [6]"*

*".mouse LC is small (about 500 μm in sagittal plane) [11] and lacks obvious localization landmarks"*

*"the intact brains of adult mice were placed on a horizontal surface and vertically cut along the posterior margin of the bilateral cerebral cortex. In the coronal cutting plane, the aqueduct of midbrain can be seen easily with the naked eyes. After embedding the cerebellum part with optimal cutting temperature (OCT) compound, coronal brain slices were cut from the cutting plane, within 1 mm, the aqueduct of midbrain disappeared and the fourth ventricle appeared, then the brain slices contained LC and were collected. From the first collection, at ~200 μm, the noradrenergic neurons' most enriched brain slices can be collected. The tyrosine hydroxylase immunofluorescence staining confirmed that the localization of LC with this method is accurate and the noradrenergic neuron most abundant slices can be determined with this method."*

### (Gargano et al., 2020)

Lack of Cannabinoid Receptor Type-1 Leads to Enhanced Age-Related Neuronal Loss in the Locus Coeruleus

*"The locus coeruleus (LC), or "blue spot", is a small nucleus located in the pons of the brainstem, lateral to the IVth ventricle. It projects throughout the neuroaxis and represents the major source of noradrenaline (NE) in the central nervous system. Although the LC consists of a relatively low number of noradrenergic neurons (45-50 thousand in a normal healthy young adult human*

*(Sharma et al., 2010) and approximately 1500 in adult mice (Sturrock et al., 1985)), its projecting area is extremely wide; noradrenergic terminals are abundantly present in the cortex, mainly in the somatosensory and motor cortex (Gaspar et al., 1989), hippocampus and amygdala (Gaspar et al., 1989)(Sadikot et al., 1990)(Radley et al., 2008), hypothalamus (Ginsberg et al., 1993) and in the brainstem itself (Westlund et al., 1980). The LC also receives a large number of afferents, mainly from the cortex, amygdala [8] (Pammer et al., 1990) and from the spinal cord (Westlund et al., 1996)"*

*".Electrophysiological studies revealed that LC noradrenergic neurons have a peculiar double firing mode; they are able to switch between tonic or phasic mode, regulating different behavioral states. A tonic firing is related to a "low attention" mode, while the switch to the phasic firing happens in response to a relevant "focus-demanding" stimuli (Aston-Jones et al., 2005). These studies suggested that LC may play a role in modulating numerous cognitive functions, including attention, memory as well as in the regulation of sleep-wake states and stress response (Benarroch, 2017). Under physiological conditions, LC neurons respond to external and internal sensory stimuli and this response influences learning and memory processes (Uematsu et al., 2015)."*

**(Kelly et al., 2023)**

Subunit-specific expression and function of AMPA receptors in the mouse locus coeruleus

*"The locus coeruleus (LC) is a collection of neurons, located bilaterally within the dorsal pons, adjacent to the fourth ventricle (Dahlstroem et al., 1964). LC neurons are distinguished from surrounding structures by their production of the neurotransmitter noradrenaline (NA), which in turn facilitates the brain-wide modulatory role of this nucleus (Berridge & Waterhouse, 2003). By projecting to almost the entire brain, the LC-NA system modulates ongoing brain activity, thereby influencing some of the most salient brain functions, including cognition (Usher et al., 1999), memory (Sara, 2015), arousal (Carter et al., 2010)), mood (McCall et al., 2015) and the central limb of the stress response (Valentino et al., 2008). In addition, the LC also modulates the physiology of various peripheral systems including the control of respiration (Carvalho et al., 2014)(Gargaglioni et al., 2010) and cardiac function (Biancardi et al., 2014)(Patrone et al., 2014)(Yao et al., 2005)"*

*".Central to LC function, in both health and disease, is the spontaneous firing rate (FR) of the principal neurons which determine the pattern and amount of NA release"*

*".In the awake state and in in vitro preparations, LC neurons characteristically fire action potentials spontaneously, with a frequency ranging from ~1 to 5 Hz (Berridge & Waterhouse, 2003)"*

*".GluA1 immunoreactive clusters were associated with puncta immunoreactive for VGLUT2 on somata, and VGLUT1 on distal dendrites. GluA4 was associated with these synaptic markers only in the distal dendrites. No specific signal was detected for the GluA2-3 subunits."*

**(Lee et al., 2024)**

Sex Differences in Single Neuron Function and Proteomics Profiles Examined by Patch-Clamp and Mass Spectrometry in the Locus Coeruleus of the Adult Mouse

*"A1 noradrenergic neurons in the medulla, A2 neurons in the nucleus tractus solitarius and A6 neurons in the LC project to many areas in the CNS, including the cortex, hippocampus, thalamus, and spinal cord 18 . The LC, a small nucleus located on either side of the fourth ventricle, contains the largest cluster of noradrenergic neurons in the CNS. LC neurons project to the cortex, hippocampus, amygdala, hypothalamus, cerebellum, and spinal cord 18 and modulate physiological processes, such as arousal, attention, cognition, anxiety, and responses to stress 18-21"*

*"We identified LC noradrenergic neurons in the pons by their location along the lateral margin of the fourth ventricle 28, 29 and by expression of tyrosine hydroxylase (TH) 26, a key enzyme involved in catecholamine synthesis. We also labeled LC neurons with anti-dopamine beta-hydroxylase (DBH) antibodies, an enzyme that converts dopamine to noradrenaline and is a marker of noradrenergic neurons 26, 30."*

**(Zhou et al., 2024)**

Progressive noradrenergic degeneration and motor cortical dysfunction in Parkinson's disease

*"Locus coeruleus (LC) is a bilateral structure situated in the upper dorsolateral pontine tegmentum. LC neurons synthesize and release norepinephrine (NE, also known as noradrenaline), which is essential for regulating brain function. LC of both hemispheres comprises about 50,000 cells in humans and around 3000 cells in rodents [1](Swanson, 1976)(Baker et al., 2004)(German et al., 1988). The LC-NE neurons project to broad brain regions (e.g., the cerebral cortex, the thalamus, the amygdala, the hippocampus, the cerebellum, and the medulla) and play a pivotal role in regulating normal brain function and behavioral states, including arousal, attention, cognition, and stress responses. Noradrenergic receptors (ARs) can be categorized into 3 subtypes, including α1, α2, and β receptors, which are G-protein coupled receptors (GPCRs). While α1 and β receptors are coupled with G q and G s proteins, respectively, α2 receptors are coupled with G i protein and the subsequent signaling cascades (Perez, 2020). Particularly, α2 receptors are expressed highly at noradrenergic axon terminals, where they can act as autoreceptors to regulate presynaptic NE release. Thus, NE has a higher affinity for α2 receptors than α1 and β receptors."*

**(Wagner-Altendorf et al., 2019)**

Axonal projection-specific differences in somatodendritic α2 autoreceptor function in locus coeruleus neurons

*"The pontine nucleus locus coeruleus (LC) contains the majority-about 10,000-15,000 neurons on each side in humans and 1,500 in rodents-of central noradrenergic neurons that project to a wide variety of targets including cerebellum, prefrontal cortex and hippocampus (Berridge et al., 2003)(Moore et al., 1979)(Swanson, 1976)"*

*".Electrophysiological studies of LC neurons have identified several characteristic functional features such as low-frequency pacemaker activity with broad action potentials, transient high-frequency burst discharges in response to salient stimuli and an apparently homogeneous inhibition of firing by activation of somatodendritic α2 autoreceptors (α2AR)"*

*".Using fluorescent beads-mediated retrograde tracing in adult C57Bl6/N mice, we compared the anatomical distributions and functional in vitro properties of identified LC neurons projecting either to medial prefrontal cortex, hippocampus or cerebellum. The functional in vitro analysis of LC neurons confirmed their mostly uniform functional properties regarding action potential generation and pacemaker firing. However, we identified significant differences in tonic and evoked α2AR-mediated responses. While hippocampal-projecting LC neurons were partially inhibited by endogenous levels of norepinephrine and almost completely silenced by application of saturating concentrations of the α2 agonist clonidine, prefrontal-projecting LC neurons were not affected by endogenous levels of norepinephrine and only partially inhibited by saturating concentrations of clonidine."*

(Carter et al., 2010)

Tuning arousal with optogenetic modulation of locus coeruleus neurons

*"Neural activity in the noradrenergic locus coeruleus correlates with periods of wakefulness and arousal. However, it is unclear whether tonic or phasic activity in these neurons is necessary or sufficient to induce transitions between behavioral states and to promote long-term arousal. Using optogenetic tools in mice, we found that there is a frequency-dependent, causal relationship among locus coeruleus firing, cortical activity, sleep-to-wake transitions and general locomotor arousal. We also found that sustained, high-frequency stimulation of the locus coeruleus at frequencies of 5 Hz and above caused reversible behavioral arrests. These results suggest that the locus coeruleus is finely tuned to regulate organismal arousal and that bursts of noradrenergic overexcitation cause behavioral attacks that resemble those seen in people with neuropsychiatric disorders."*

(McCall et al., 2015)

CRH engagement of the locus coeruleus noradrenergic system mediates stress-induced anxiety

*"Summary The locus coeruleus noradrenergic (LC-NE) system is one of the first systems engaged following a stressful event. While numerous groups have demonstrated that LC-NE neurons are activated by many different stressors, the underlying neural circuitry and the role of this activity in generating stress-induced anxiety has not been elucidated. Using a combination of in vivo chemogenetics, optogenetics, and retrograde tracing we determine that increased tonic activity of the LC-NE system is necessary and sufficient for stress-induced anxiety and aversion. Selective inhibition of LC-NE neurons during stress prevents subsequent anxiety-like behavior. Exogenously increasing tonic, but not phasic, activity of LC-NE neurons is alone sufficient for anxiety-like and aversive behavior. Furthermore, endogenous corticotropin releasing hormone+ (CRH+) LC inputs from the amygdala increase tonic LC activity, inducing anxiety-like behaviors. These studies position the LC-NE system as a critical mediator of acute stress-induced anxiety and offer a potential intervention for preventing stress-related affective disorders."*

(Uematsu et al., 2015)

Projection specificity in heterogeneous locus coeruleus cell populations: implications for learning and memory

*"Noradrenergic neurons in the locus coeruleus (LC) play a critical role in many functions including learning and memory. This relatively small population of cells sends widespread projections throughout the brain including to a number of regions such as the amygdala which is involved in emotional associative learning and the medial prefrontal cortex which is important for facilitating flexibility when learning rules change"*

*".Previous work has demonstrated that subpopulations of LC noradrenergic cells innervate specific brain regions suggesting heterogeneity of function in LC neurons."*

(Takeuchi et al., 2016)

Locus coeruleus and dopaminergic consolidation of everyday memory

*"The retention of episodic-like memory is enhanced, in humans and animals, when something novel happens shortly before or after encoding. Using an everyday memory task in mice, we sought the neurons mediating this dopamine-dependent novelty effect, previously thought to originate exclusively from the tyrosine-hydroxylase-expressing (TH+) neurons in the ventral tegmental area. Here we report that neuronal firing in the locus coeruleus is especially sensitive to environmental novelty, locus coeruleus TH+ neurons project more profusely than ventral tegmental area TH+ neurons to the hippocampus, optogenetic activation of locus coeruleus TH+ neurons mimics the novelty effect, and this novelty-associated memory enhancement is unaffected by ventral tegmental area inactivation. Surprisingly, two effects of locus coeruleus TH+ photoactivation are sensitive to hippocampal D1/D5 receptor blockade and resistant to adrenoceptor blockade: memory enhancement and long-lasting potentiation of synaptic transmission in CA1 ex vivo. Thus, locus coeruleus TH+ neurons can mediate post-encoding memory enhancement in a manner*

*consistent with possible co-release of dopamine in the hippocampus."*

### (Breton-Provencher et al., 2022)

Spatiotemporal dynamics of noradrenaline during learned behavior

*"Noradrenaline released from the locus coeruleus (LC) is a ubiquitous neuromodulator1–4 that has been linked to multiple functions including arousal5–8, action and sensory gain9–11, and learning12–16"*

*".Here we show that LC-NA activity displays distinct spatiotemporal dynamics to enable two functions during learned behaviour: facilitating task execution and encoding reinforcement to improve performance accuracy"*

*".Targeted recordings of LC-NA neurons using photo-tagging, two-photon micro-endoscopy and two-photon output monitoring showed that transient LC-NA activation preceded behavioural execution and followed reinforcement. These two components of phasic activity were heterogeneously represented in LC-NA cortical outputs, such that the behavioural response signal was higher in the motor cortex and facilitated task execution, whereas the negative reinforcement signal was widely distributed among cortical regions and improved response sensitivity on the subsequent trial. Modular targeting of LC outputs thus enables diverse functions, whereby some noradrenaline signals are segregated among targets, whereas others are broadly distributed. Noradrenaline-expressing neurons in the locus coeruleus in mouse facilitate task execution and encode reinforcement in learning tasks, via partially modular projections to the cortex."*

### (Su et al., 2026)

Topographic structure and function of locus coeruleus norepinephrine neurons

*"Norepinephrine (NE) is released throughout most of the central nervous system by neurons in the locus coeruleus (LC). We found a relationship between the morphologies, gene expression, and activity of LC-NE neurons in mice. Axonal projections of individual neurons were extensive but largely confined to subsets of brain regions. Axonal projections and graded gene expression correlated with locations of cell bodies in LC"*

*".neurons in dorsal LC projecting to the cerebral cortex were excited when mice made a different choice from the previous one and by reward prediction errors, a signal driving learning. Background activity of neurons in ventral LC was higher when mice ignored stimuli indicating potential reward availability."*

### (Su et al._1, 2026)

Topographic structure and function of locus coeruleus noradrenaline neurons.

*"Noradrenaline (also known as norepinephrine) is released throughout most of the central nervous system by neurons in the locus coeruleus1-4. Here we identified a relationship between the morphologies, gene expression and activity of noradrenergic neurons in locus coeruleus in mice. Axonal projections of individual neurons were extensive but largely confined to subsets of brain regions. Axonal projections and graded gene expression correlated with locations of cell bodies in locus coeruleus. In a behavioural task requiring continuous learning from actions, dorsal locus coeruleus neurons projecting to isocortex were activated when mice switched choices and by reward-prediction-error signals that drive learning. Background activity of neurons in ventral locus coeruleus was higher when mice ignored stimuli indicating potential reward availability."*

### (Alam et al., 2023)

Electrophysiology as a Tool to Decipher the Network Mechanism of Visceral Pain in Functional Gastrointestinal Disorders

*"The brain nucleus locus coeruleus (LC) is located in the dorsolateral pons and is a significant source of norepinephrine (NE) in the brain (Figure 2). Retrograde tracing studies in rats demonstrated that ascending axons of noradrenergic LC neurons project to specific regions of the CNS including the thalamus, mPFC, ACC, hippocampus, hypothalamus, amygdala, and cerebellum (Berridge et al., 2003)(Mason et al., 1979), while descending axons project to the spinal dorsal horn (SDH) (Proudfit et al., 1991). The LC releases NE to modulate nociception by altering the excitability of spinothalamic tract neurons [131](Treviño et al., 1972)(Willis et al., 1974). These data suggest that LC and sub-coeruleus (SC) regions provide noradrenergic innervation to the forebrain, cerebellum, brainstem, and dorsal and ventral horns of the spinal cord (Proudfit et al., 1991)(Clark et al., 1991)(Sluka et al., 1992). Electrical stimulation of LC neurons produces antinociception (Janss et al., 1987)(Jones et al., 1986)(Jones et al., 1986)"*

*"A recent study used chemogenetic, the Designer Receptors Exclusively Activated by Designer Drugs (DREADD), to activate the LC-SC pathway selectively and explore its effect on neuropathic pain in mice. Chemogenetic activation of the LC-SC pathway increased the release of NE in the SDH and reduced pain intensity in neuropathic mice (Li et al., 2022)."*

### (Li et al., 2022)

Activation of locus coeruleus-spinal cord noradrenergic neurons alleviates neuropathic pain in mice via reducing neuroinflammation from astrocytes and microglia in spinal dorsal horn

*"Background The noradrenergic neurons of locus coeruleus (LC) project to the spinal dorsal horn (SDH), and release norepinephrine (NE) to inhibit pain transmission. However, its effect on pathological pain and the cellular mechanism in the SDH remains unclear.*

*This study aimed to explore the analgesic effects and the anti-neuroinflammation mechanism of LC-spinal cord noradrenergic pathway (LC:SC) in neuropathic pain (NP) mice with sciatic chronic constriction injury. Methods The Designer Receptors Exclusively Activated by Designer Drugs (DREADD) was used to selectively activate LC:SC. Noradrenergic neuron-specific retro–adeno-associated virus was injected to the spinal cord. Pain threshold, LC and wide dynamic range (WDR) neuron firing, neuroinflammation (microglia and astrocyte activation, cytokine expression), and α2AR expression in SDH were evaluated. Results Activation of LC:SC with DREADD increased the mechanical and thermal nociceptive thresholds and reduced the WDR neuron firing. LC:SC activation (daily, 7 days) downregulated TNF-α and IL-1β expression, upregulated IL-4 and IL-10 expression in SDH, and inhibited microglia and astrocytes activation in NP mice. Immunofluorescence double staining confirmed that LC:SC activation decreased the expression of cytokines in microglia of the SDH. In addition, the effects of LC:SC activation could be reversed by intrathecal injection of yohimbine. Immunofluorescence of SDH showed that NE receptor α2B-AR was highly expressed in microglia in CCI mice. Conclusion These findings indicate that selective activation of LC:SC alleviates NP in mice by increasing the release of NE and reducing neuroinflammation of astrocytes and microglia in SDH."*

**(Dimitrov et al., 2013)**

Forebrain GABAergic projections to locus coeruleus in mouse

*"The noradrenergic locus coeruleus (LC) regulates arousal, memory, sympathetic nervous system activity, and pain"*

*". CAmy and PLH projections were densest in a pericoerulear zone, but many fibers entered the LC proper. Close apposition between CAmy GABAergic projections and TH-ir processes suggests that CAmy GABAergic neurons may directly inhibit noradrenergic principal neurons."*

**(Gugel et al., 2024)**

LC-derived excitatory synaptic transmission to dorsal raphe serotonin neurons is inhibited by activation of alpha2-adrenergic receptors

*"Noradrenaline afferents in the dorsal raphe arise from the locus coeruleus system"*

*"Initially, we performed retrograde tracing of Fluoro-Gold™ (FG, (Saleeba et al., 2019), n = 3 mice) microinjected in the dorsal raphe to the noradrenergic brain nuclei; broadly divided as the locus coeruleus (LC) system (as defined in prior work (Grzanna et al., 1980)[18] [19], Fig. 1A) and non-LC nuclei in the ventral and dorsomedial medulla"*

*".Noradrenaline neurons were identified by location and immunostaining for tyrosine hydroxylase (TH), the rate-limiting enzyme in the production of dopamine and noradrenaline. On average, FG was found in 94% of TH + neurons in LC proper (555/593 neurons, Fig. S1A), 70% of TH + neurons in sub-coeruleus (Sub-C, 334/479 neurons, Fig. S1B) and 70% of TH + neurons in the rostral non-contiguous extension of LC (Rnc-LC, 95/137 neurons, Fig. S1C [17],). In contrast, 18% of TH + neurons in the medulla contained FG (16/90 neurons, Fig. S1D). To test for functional connectivity between the LC system and the dorsal raphe, an optogenetic strategy was employed using the TH-Cre transgenic mouse line [14] that expresses Cre-recombinase in TH + (dopamine and noradrenaline) neurons"*

*". we found that selective optogenetic activation of locus coeruleus terminals in the dorsal raphe was sufficient to produce an α1-AR-mediated excitatory postsynaptic current (α1-AR-EPSC). Activation of inhibitory α2-adrenergic receptors (α2-AR) with UK-14,304 eliminated the α1-AR-EPSC via presynaptic inhibition of noradrenaline release, likely via inhibition of voltage-gated calcium channels. In a subset of serotonin neurons, activation of postsynaptic α2-AR produced an outward current through activation of GIRK potassium conductance. Further, in vivo activation of α2-AR by systemic administration of clonidine reduced the expression of c-fos in the dorsal raphe serotonin neurons, indicating reduced neural activity."*

**(Geng et al., 2025)**

Noradrenergic inputs from the locus coeruleus to anterior piriform cortex and the olfactory bulb modulate olfactory outputs

*"In DBH-Cre mice model, we show that LC NAergic neurons project directly to both anterior piriform cortex (aPC) and the olfactory bulb (OB)."*

# Developmental origin and subpopulation organization

*Most mouse LC noradrenergic neurons come from an embryonic En1-positive progenitor domain in the rostral hindbrain, with strong evidence that the LC is split into developmental subgroups rather than being one uniform lineage. These lineage differences map onto dorsal/ventral position, nearby subcoeruleus/A7-related cells, and distinct output patterns, helping explain adult LC heterogeneity. (5 sources)*

Developmentally, the mouse LC is now best understood as a **lineage-structured noradrenergic population** rather than a single homogeneous nucleus. A key point is that **>99% of LC noradrenergic neurons derive from the embryonic hindbrain progenitor domain marked by En1**, making En1 lineage a near-universal

developmental origin for the LC proper. (Chandler et al., 2019) (Robertson et al., 2013) This En1-derived population belongs to the broader rostral hindbrain noradrenergic system, and developmental mapping places these neurons across **rhombomeres 1–5**, with the LC itself arising mainly from the **isthmus/r0 and r1 region**. (Manger et al., 2021) (Robertson et al., 2013) (Plummer et al., 2017)

Within that En1-derived system, the LC is further subdivided by developmental origin. Fate-mapping work indicates that the adult LC contains **r0-derived dorsal** and **r1-derived ventral** components, so even the compact LC core is developmentally partitioned along the dorsoventral axis. (Plummer et al., 2017) This fits with evidence that the LC complex forms a **continuum** extending from the compact LC proper into more dispersed **subcoeruleus and A7-related noradrenergic cells**, rather than having a perfectly sharp boundary between these groups. (Plummer et al., 2017) (Manger et al., 2021)

Development also helps explain adult cell-type diversity. The LC has long been treated as the A6 noradrenergic group, but lineage studies identified **four genetically separable subpopulations of mature norepinephrine neurons** in the developing/mature mouse hindbrain, each with distinct anatomical distribution, axon morphology, and projection pattern. (Robertson et al., 2013) (Chandler et al., 2019) One important result from this work was that some cortical norepinephrine input arises from neurons **outside the classic LC**, including projections to prefrontal/orbitofrontal and insular cortex, which argues against a simple "all cortical NE comes from LC proper" model. (Chandler et al., 2019) (Robertson et al., 2013)

A second developmental split comes from **alar plate versus non-alar origin**. Most of the LC proper appears to arise from the **Pax7-defined alar plate**, consistent with a dorsal neural tube origin followed by migration to its adult position. (Plummer et al., 2017) But this is not the whole story: a **smaller Pax7-negative noradrenergic subpopulation** is also present across the dorsoventral LC complex, including subcoeruleus/A7-related regions, and this group shows **very sparse thalamic projections** compared with the broader LC complex. (Plummer et al., 2017) This suggests that developmental lineage is tied not just to where neurons end up, but also to what targets they innervate.

At the systems level, this means the adult mouse LC should be viewed as a **developmentally assembled complex**: a dense A6 core embedded in a larger rostral hindbrain noradrenergic field that includes nearby SubC/A7 elements and peri-LC cell populations. (Manger et al., 2021) Recent work combining tracing with single-cell and spatial transcriptomic approaches reinforces this picture by showing that both **LC noradrenergic neurons and peri-LC neurons are anatomically, functionally, and transcriptionally heterogeneous**. (Luskin et al., 2025) In practice, developmental origin, adult position, and projection target all appear to align only partially, producing the mixed but structured subpopulation organization seen in the mouse LC today.

## Evidence

**(Chandler et al., 2019)**

Redefining Noradrenergic Neuromodulation of Behavior: Impacts of a Modular Locus Coeruleus Architecture

> *"Using this approach, four subsets of neurons were identified, each distinct in their anatomical distribution and efferent projection pattern. In addition, an unexpected projection to the orbital frontal cortex and insular cortex was found arising from outside of the LC, contradicting the dogmatic view that the LC is the lone noradrenergic nucleus projecting to the cortex (Robertson et al., 2013)"*

> *".Greater than 99% of LC-NE neurons are derived from the embryonic hindbrain progenitor domain defined by En1 expression."*

**(Robertson et al., 2013)**

Developmental origins of central norepinephrine neuron diversity

> *"Central norepinephrine-producing neurons comprise a diverse population of cells differing in anatomical location, connectivity, function and response to disease and environmental insult. The mechanisms that generate this diversity are unknown. Here we elucidate the lineal relationship between molecularly distinct progenitor populations in the developing mouse hindbrain and mature norepinephrine neuron subtype identity. We have identified four genetically separable subpopulations of mature norepinephrine neurons differing in their anatomical location, axon morphology and efferent projection pattern. One of the subpopulations showed*

*an unexpected projection to the prefrontal cortex, challenging the long-held belief that the locus coeruleus is the sole source of norepinephrine projections to the cortex. These findings reveal the embryonic origins of central norepinephrine neurons and provide multiple molecular points of entry for future study of individual norepinephrine circuits in complex behavioral and physiological processes including arousal, attention, mood, memory, appetite and homeostasis."*

### (Manger et al., 2021)

The Mammalian Locus Coeruleus Complex—Consistencies and Variances in Nuclear Organization

*"These noradrenergic neurons have been shown to be derived from rhombomeres 1-5 in the developing mouse brain (Robertson et al., 2013)"*

*"in mammals, with one known exception (Imam et al., 2018), the noradrenergic neurons of the locus coeruleus complex are found within the periventricular grey matter and the parvicellular reticular nucleus of the rostral hindbrain"*

*"the locus coeruleus complex is comprised of several nuclei"*

*"Descriptions of the nuclear parcellation of the locus coeruleus complex have been provided in approximately 80 mammal species spanning the phylogenetic breadth of this class. Within the mammalian rostral hindbrain, noradrenergic neurons (revealed with tyrosine hydroxylase and dopamine-ß-hydroxylase immunohistochemistry) have been observed within the periventricular grey matter (A4 and A6 nuclei) and parvicellular reticular nucleus (A5 and A7 nuclei)"*

*"In most mammals, the A6 is comprised of a moderate density of neurons, but in Murid rodents, primates, and megachiropteran bats, the A6 exhibits a very high density of neurons."*

### (Plummer et al., 2017)

Two Subpopulations of Noradrenergic Neurons in the Locus Coeruleus Complex Distinguished by Expression of the Dorsal Neural Tube Marker Pax7

*"The population arising from the isthmus (r0) and r1, as defined by early expression of En1, comprises the majority of the locus coeruleus (LC), the largest anatomically defined noradrenergic population and a major source of norepinephrine released in the cortex, as well as part of the dorsal subcoeruleus (SubC) and A7 noradrenergic nuclei in the pons. More recently, fate mapping on the basis of Fgf8 Cre expression in r0 has demonstrated that the LC is divided between r0-and r1-derived subpopulations (Watson et al., 2017). A dorsal subpopulation is derived from the Fgf8 Cre expression domain encompassing r0, and by inference, the remaining ventral subpopulation is derived from r1"*

*".The developmental progression of Phox2a and Phox2b expression during LC development suggests that the neurons of the LC originate in the alar plate, the dorsal subdivision of the neural tube, and migrate to their final position in the basal plate (Aroca et al., 2006). In chick, the alar origin of the LC was confirmed experimentally by fate mapping with quail/chick chimeras and immunohistochemical analysis using Pax7 as a marker of the alar plate (Aroca et al., 2006)"*

*".Using intersectional genetic fate-mapping, we have previously shown that noradrenergic neurons derived from the En1 expression domain of the embryonic hindbrain populate 99.8% of the locus coeruleus (LC) and a portion of the dorsal subcoeruleus and A7 nuclei (Robertson et al., 2013) according to an adult mouse brain atlas (Paxinos and Franklin, 2013). These neurons (hereafter designated the LC complex) form a continuum extending from the compact LC proper to the more dispersed cells of the SubC and A7 nuclei (Plummer et al., 2015)"*

*".we confirm that the majority of the LC proper arises from the alar plate, the dorsal domain of the neural tube, as defined by expression of Pax7Cre. In addition, our analysis shows that a large proportion of the En1-derived A7 and SubC nuclei also originate in the Pax7Cre-defined alar plate. Surprisingly, however, we identify a smaller subpopulation of the LC complex that arises from outside the Pax7Cre expression domain. We characterize the distribution of these neurons within the LC complex, their cell morphology, and their axonal projection pattern. Compared to the broader LC complex, the newly identified Pax7Cre-negative noradrenergic subpopulation has very sparse projections to thalamic nuclei, suggestive of distinct functions"*

*".The projection patterns of the two Pax7 Cre -defined subpopulations of the LC complex may point to functional differences; the virtual absence of projections to the thalamus suggests that the Pax7 Cre -negative subpopulation does not directly participate in the modulation of thalamic neuron activity performed by the broader LC complex (Berridge and Waterhouse, 2003)"*

*".We observe Pax7 Crenegative noradrenergic neurons throughout the dorsoventral extent of the adult LC complex, suggesting that they originate in both r0 and r1."*

### (Luskin et al., 2025)

Heterogeneous pericoerulear neurons tune arousal and exploratory behaviors

*"As the primary source of norepinephrine (NE) in the brain, the locus coeruleus (LC) regulates arousal, avoidance and stress responses1,2"*

*". We define peri-LC anatomy using viral tracing and combine single-cell RNA sequencing with spatial transcriptomics to molecularly define both LCNE and peri-LC cell types. We identify several neuronal cell types which underlie peri-LC functional diversity using a series of complementary neural circuit approaches in behaving mice. Our findings indicate that LC and peri-LC neurons are transcriptionally, functionally, and anatomically heterogenous neuronal populations which modulate arousal and avoidance states."*

## Transcriptomics datasets and molecular profiling

*Transcriptomic work in mouse LC shows that noradrenergic neurons share a strong core NE gene program but are not all the same. Available datasets include bulk/translational profiling, single-cell or single-nucleus RNA-seq, and newer spatial transcriptomics, which together show neuropeptide-rich molecular diversity and graded subtypes linked to position and projections. (7 sources)*

- **Bulk/translational profiling of mouse LC noradrenergic neurons**
- A notable molecular profiling study used **bacTRAP translational profiling** in **Th-EGFP-L10a mice** to isolate ribosome-bound mRNAs from mouse LC noradrenergic neurons under baseline conditions. This dataset identified **3,045 genes significantly enriched** in LC neurons and defined **over 3,000 high-confidence transcripts**, including many potentially druggable receptors. It also confirmed that the transgene labeled **100% of TH+ / DBH+ LC neurons** in their material. (Mulvey et al., 2018)
- This study is especially useful for marker selection because it found strong enrichment of the canonical noradrenergic machinery, including **Th, Dbh, Slc6a2 (NET), Slc18a2 (VMAT2), and Gch1**. (Mulvey et al., 2018)
- **Dataset/accession:** not provided in the supplied quote. (Mulvey et al., 2018)
- **Single-cell RNA-seq focused on neuropeptides in LC NE neurons**
- A later study combined **single-cell RNA sequencing, neuroanatomy, and electrophysiology** to map **galanin and its receptors** in the mouse LC. It showed that LC NE neurons express **galanin**, **more than 19 other neuropeptides**, and **more than 30 neuropeptide receptors**, with **Galr1** present at low levels. It also reported that **Npy was not expressed in LC NE neurons**, supporting the idea that NPY is more a peri-LC than LC-NE marker in mouse. (Caramia et al., 2023)
- This dataset is useful because it expands LC identity beyond the core catecholamine genes and shows that mouse LC neurons are **neuropeptidergically rich**, with molecular diversity likely tied to differences in local regulation. (Caramia et al., 2023) (Becker et al., 2025)
- **Dataset/accession:** not provided in the supplied quote. (Caramia et al., 2023)
- **Single-cell RNA-seq plus spatial transcriptomics across LC and peri-LC**
- Newer work combined **single-cell RNA sequencing with spatial transcriptomics** to define both **LC noradrenergic (LCNE)** and **peri-LC** cell types. This study concluded that **LC and peri-LC neurons are transcriptionally, functionally, and anatomically heterogeneous**, and it identified several neuronal cell types that likely underlie peri-LC functional diversity. (Luskin et al., 2025)
- This kind of dataset is especially helpful for mouse LC because the nucleus is small and embedded in a heterogeneous surround; spatial mapping helps separate the **dense NE core** from neighboring non-NE populations. (Luskin et al., 2025)
- **Dataset/accession:** not provided in the supplied quote. (Luskin et al., 2025)
- **Single-nucleus RNA-seq plus spatial transcriptomics in normal and injured brainstem**
- Another study used **single-nucleus RNA-seq (snRNA-seq)** together with **spatial transcriptome RNA-seq** in **normal and mild traumatic brain injury (mTBI) mouse brainstem**. It resolved **35 neuron and non-neuron clusters** and reported enhanced communication among **noradrenergic neurons in the LC**, with some LC-

related interactions being especially injury-sensitive. (Zhuang et al., 2025)

- Although this was a broader brainstem dataset rather than an LC-only atlas, it is relevant for LC biology because it places LC noradrenergic neurons into a wider spatial and cell-cell communication framework and shows how their transcriptional state may change after injury. (Zhuang et al., 2025)
- **Dataset/accession:** not provided in the supplied quote. (Zhuang et al., 2025)
- **Gene-expression gradients linked to morphology, position, and output**
- Recent mouse work reported a relationship between **gene expression, morphology, and activity** in LC-NE neurons. In that study, **graded gene expression** correlated with the **position of cell bodies within the LC**, and this in turn matched biased projection patterns and distinct task-related activity. (Su et al., 2026) (Su et al._1, 2026)
- This supports a useful transcriptomic conclusion: mouse LC neurons do not appear to split only into a few hard classes, but can also vary along **spatial molecular gradients** that relate to dorsal-versus-ventral location and projection bias. (Su et al., 2026) (Su et al._1, 2026)
- **Review-level summary of the molecular picture**
- A recent review emphasizes that newer LC transcriptional profiling has revealed a broad **neuropeptidergic landscape** in and around the LC, and that these data offer an important window into LC neuronal diversity. (Becker et al., 2025)
- Taken together, the transcriptomic literature supports a model in which mouse LC noradrenergic neurons share a common **NE synthesis/transport program** but vary in **neuropeptide content, receptor expression, spatial position, and relation to peri-LC cell types**. (Mulvey et al., 2018) (Caramia et al., 2023) (Luskin et al., 2025) (Su et al., 2026)
- **Dataset accession note**
- For the studies quoted here, **dataset accession numbers were not included in the provided excerpts**, so they cannot be listed reliably from the supplied material. If needed, these should be checked directly in the original papers or their GEO/SRA supplementary information. (Mulvey et al., 2018) (Caramia et al., 2023) (Luskin et al., 2025) (Zhuang et al., 2025)

## Evidence

**(Mulvey et al., 2018)**

Molecular and Functional Sex Differences of Noradrenergic Neurons in the Mouse Locus Coeruleus

*"The locus coeruleus (LC) is the largest source of forebrain norepinephrine (NE), and abnormalities in LC-NE function contribute to diseases such as anxiety, depression, post-traumatic stress disorder (PTSD), and attention deficit hyperactive disorder (ADHD)"*

*".In the mouse, the LC is comprised of approximately 2,000 neurons per side, the majority of which are uniquely noradrenergic (Robertson et al., 2013; Chandler et al., 2014)"*

*".We profiled mouse noradrenergic neurons in vivo, defining over 3,000 high-confidence transcripts expressed therein, including druggable receptors"*

*".To compare molecular differences in female and male LC neurons, we developed a translational profiling strategy by combining the bacTRAP technology with a tyrosine hydroxylase driver line, generating Th-EGFP-L10a transgenic mice"*

*".we observed GFP-positive cells in catecholaminergic populations throughout the brain and retina, including in the LC, ventral tegmental area (VTA), substantia nigra pars compacta (SNc), hypothalamic arcuate nucleus, and olfactory bulb, among others, and in retinal amacrine and ganglion cells"*

*".We additionally stained transgenic LC sections for DBH, and we observed 100% co-expression of the EGFP-L10a transgene in TH-*

*positive, DBH-positive LC neurons (Figure 1A)"*

*".We then used this line to molecularly profile female and male LC neurons under baseline conditions"*

*".We generated four biological replicates each of female and male adult LC neuron input and immunoprecipitation (IP) samples"*

*".In total, we identified 3,045 genes significantly enriched in LC neurons, and these represent a molecular profile of LC neurons in the mouse"*

*".We note that genes required for NE synthesis and release, including Gch1, Slc6a2, Slc18a2, Dbh, and Th, are highly enriched in the LC translatome, and therefore constitute useful markers for this neuronal population."*

(Caramia et al., 2023)

Neuronal diversity of neuropeptide signaling, including galanin, in the mouse locus coeruleus

*"The locus coeruleus (LC) is the main source of norepinephrine (NE) in the brain, modulates wakefulness, and is altered in stress-related disorders"*

*".Here, we combine single-cell-RNA-sequencing, neuroanatomy, and electrophysiology to build a map of galanin and its receptors in the mouse LC. We found that NE neurons express mRNA transcripts for galanin, >19 other neuropeptides (but Npy), and >30 neuropeptide receptors with Galr1 included at low levels. Selective activation of GalR1 inhibited the spontaneous activity of a subset of NE neurons possibly through the activation of Kv currents."*

(Becker et al., 2025)

Neuropeptides as transmitters and regulators of the locus coeruleus noradrenergic system

*"The locus coeruleus (LC), the brain's main source of noradrenaline"*

*"we now understand the LC as a multiplexed nucleus, capable of temporally precise and targeted neuromodulation of distinct brain regions and functions. The LC neuropeptidergic landscape provides a window into this remarkable neuronal diversity. Stemming from recent technological advances that have allowed for LC transcriptional profiling, a wealth of data on the (co)expression of LC neuropeptides and their cognate receptors has come to light"*

*"Here we compile existing literature on the expression, anatomical distribution, physiological effects, and, when available, behavioral role of the major neuropeptidergic populations of, and innervating, the LC and peri-LC."*

(Luskin et al., 2025)

Heterogeneous pericoerulear neurons tune arousal and exploratory behaviors

*"As the primary source of norepinephrine (NE) in the brain, the locus coeruleus (LC) regulates arousal, avoidance and stress responses1,2"*

*". We define peri-LC anatomy using viral tracing and combine single-cell RNA sequencing with spatial transcriptomics to molecularly define both LCNE and peri-LC cell types. We identify several neuronal cell types which underlie peri-LC functional diversity using a series of complementary neural circuit approaches in behaving mice. Our findings indicate that LC and peri-LC neurons are transcriptionally, functionally, and anatomically heterogenous neuronal populations which modulate arousal and avoidance states."*

(Zhuang et al., 2025)

Single-nucleus and spatial signatures of the brainstem in normal brain and mild traumatic brain injury in male mice

*"we performed single-nucleus RNA sequencing with spatial transcriptome sequencing in both normal and mTBI brainstems in male mice, revealing thirty-five neuron and non-neuron clusters. Typically, we identified subtypes of neurons that co-release multiple neurotransmitters, especially in the sagittal midline of the brainstem. Spatially adjacent neurons sharing similar gene expression patterns"*

*". revealed enhanced communication among the noradrenergic neurons in the locus coeruleus. Interestingly, among the interactions, the noradrenergic neurons and the Glu-GABA-Net-choline interaction, which represent the communication between the locus coeruleus and the Sol, were particularly sensitive to injury"*

*". This study utilized snRNA-seq combined with ST RNA-seq to explore the characteristics of the brainstem under both normal and injury conditions."*

(Su et al., 2026)

Topographic structure and function of locus coeruleus norepinephrine neurons

*"Norepinephrine (NE) is released throughout most of the central nervous system by neurons in the locus coeruleus (LC). We found a relationship between the morphologies, gene expression, and activity of LC-NE neurons in mice. Axonal projections of individual*

*neurons were extensive but largely confined to subsets of brain regions. Axonal projections and graded gene expression correlated with locations of cell bodies in LC"*

*".neurons in dorsal LC projecting to the cerebral cortex were excited when mice made a different choice from the previous one and by reward prediction errors, a signal driving learning. Background activity of neurons in ventral LC was higher when mice ignored stimuli indicating potential reward availability."*

(Su et al._1, 2026)

Topographic structure and function of locus coeruleus noradrenaline neurons.

*"Noradrenaline (also known as norepinephrine) is released throughout most of the central nervous system by neurons in the locus coeruleus1-4. Here we identified a relationship between the morphologies, gene expression and activity of noradrenergic neurons in locus coeruleus in mice. Axonal projections of individual neurons were extensive but largely confined to subsets of brain regions. Axonal projections and graded gene expression correlated with locations of cell bodies in locus coeruleus. In a behavioural task requiring continuous learning from actions, dorsal locus coeruleus neurons projecting to isocortex were activated when mice switched choices and by reward-prediction-error signals that drive learning. Background activity of neurons in ventral locus coeruleus was higher when mice ignored stimuli indicating potential reward availability."*